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Miami, Florida

DESIGN AND CONTROL OF A PNEUMATICALLY ACTUATED ROBOTIC MANIPULATOR FOR OFF-RISER SAMPLING AT THE HANFORD SITE

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Theophile Pierre

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To: Interim Dean Ines Triay
College of Engineering and Computing

This thesis, written by Theophile Pierre, and entitled Design and Control of a Pneumatically Actuated Robotic Manipulator for Off-Riser Sampling at the Hanford Site, having been approved in respect to style and intellectual content, is referred to you for judgment.

We have read this thesis and recommend that it be approved.

Benjamin Bosel

Dwayne McDaniel

Alicia Boymelgreen

Leonel Lagos, Major Professor

Date of Defense: November 10, 2025

The thesis of Theophile Pierre is approved.

Interim Dean Ines Triay
College of Engineering and Computing

Andrés G. Gil
Senior Vice President for Research and Economic Development
and Dean of the University Graduate School

Florida International University, 2025

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DEDICATION

To my family, whose love and sacrifices made this journey possible.

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ABSTRACT OF THE THESIS

DESIGN AND CONTROL OF A PNEUMATICALLY ACTUATED ROBOTIC MANIPULATOR FOR OFF-RISER SAMPLING AT THE HANFORD SITE

by

Theophile Pierre

Florida International University, 2025

Miami, Florida

Professor Leonel Lagos, Major Professor

The Hanford Site's plutonium production legacy presents a significant challenge: safely managing radioactive waste in deteriorating underground tanks. Traditional sampling methods are limited by access constraints, safety risks, and changing waste conditions. This research introduces a pneumatically actuated, four-degree-of-freedom robotic manipulator for off-riser sampling in these hazardous environments.

Inspired by SCARA-type robots, the manipulator navigates narrow risers and collects diverse waste forms—solid, granular, and slurry—while meeting nuclear safety standards. Its design features corrosion-resistant materials, modular tooling, and fail-safe mechanisms. Finite element analysis ensures structural reliability.

The control system uses PID algorithms to handle the nonlinear behavior of pneumatic actuators, modeled with mass flow and pressure-volume dynamics. MATLAB/Simulink and NVIDIA Issac-sim simulations and optimization strategies, including Ziegler-Nichols and Genetic Algorithms, enhance performance.

Initial prototypes demonstrated effective destructive sampling, guiding further design refinements. This system supports safer, more precise waste retrieval, aligning with the Department of Energy's nuclear remediation goals.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
Al	Aluminum (chemical element; e.g., Al-6061 alloy)
ANSI	American National Standards Institute
ASTM	ASTM International (formerly American Society for Testing and Materials)
BB	Bang-Bang
CAD	Computer-Aided Design
CNC	Computer Numerical Control
DOE-EM	Department of Energy Environmental Management
DoF	Degrees of Freedom
DST	Double-Shell Tank
E-stop	Emergency stop
FIU-ARC	Florida International University – Applied Research Center
GUI	Graphical User Interface
HORSS	Hanford Off-Riser Sampling System
HLW	High Level Waste
ISO	International Organization for Standardization
LiDAR	Light Detection and Ranging
MATLAB	Matrix Laboratory (MathWorks)
ORSS	Off-Riser Sampling System

PID	Proportional–Integral–Derivative (controller)
PETG	Polyethylene Terephthalate Glycol-modified
PMMA	Polymethyl methacrylate (acrylic)
PWM	Pulse-Width Modulation
ROS	Robot Operating System
RViz	ROS visualization environment
SST	Single-Shell Tank
TF	Transform library (ROS)
UDP	User Datagram Protocol
URDF	Unified Robot Description Format
USD	Universal Scene Description
WRPS	Washington River Protection Solutions
WTP	Waste Treatment Plant

LIST OF SYMBOLS

Symbol	Meaning
$\dot{m}_i$	Mass flow into chamber i
A_o	Orifice area (geometric)
A_0	Orifice area constant/scale used in flow model
C_d	Discharge coefficient (orifice/valve)
$\Psi(P_d, P_u)$	Compressible-flow function (piecewise, choked/unchoked)
P	Pressure (generic)
$P_u,\ P_d$	Upstream and downstream pressures across the orifice
P_{supply}	Supply pressure
P_{chamber}	Cylinder chamber pressure
P_{atm}	Atmospheric pressure
P_{cr}	Critical pressure ratio for choking
γ	Specific-heat ratio of air (≈ 1.4)
R	Specific gas constant for dry air ($\approx 287\,\text{J}/(\text{kg}\cdot\text{K})$)
T	Gas temperature (generic)
$T_u,\ T_d$	Upstream and downstream temperatures
A_c	Chamber cross-sectional area
x_{rel}	Piston position relative to charged length (incl. dead space)
m	Gas mass in chamber
$\dot{P}_i$	Time derivative of chamber i pressure
$\dot{x}$	Piston velocity
$\ddot{x}$	Piston acceleration
c_p	Specific heat at constant pressure
F_{net}	Net axial force on piston–rod assembly
δ	Direction/sign term for pressurized side (± 1)

Symbol	Meaning
$P_{\text{cap}},\ P_{\text{rod}}$	Cap-side and rod-side chamber pressures
$A_{\text{cap}},\ A_{\text{rod}}$	Cap-side and rod-side effective areas
d_b	Cylinder bore diameter
d_{piston}	Piston/rod diameter (for A_{rod})
F_{friction}	Aggregate friction force (seal/Coulomb/viscous)
F_{external}	External load (e.g., gravity/reflected payload)
m_p	Mass of piston–rod assembly
$\mu(T)$	Dynamic viscosity of air (temperature-dependent)
$\mu_0,\ T_0$	Reference dynamic viscosity and reference temperature
S	Sutherland constant for air
f	Darcy–Weisbach friction factor
Re	Reynolds number
ϵ	Pipe roughness height
d_p	Pipe inner diameter (used in Re and f)
u	Control input/valve command (e.g., voltage or duty)
$A(u)$	Orifice area as a function of command
g	Gravitational acceleration
ρ	Air density

CHAPTER 1

INTRODUCTION

1.1 Background

The production of nuclear waste is intrinsically linked to the history of nuclear technology, which began in the early twentieth century and expanded rapidly during World War II and the Cold War. This expansion culminated in the establishment of the United States Department of Energy (DOE) through the Department of Energy Organization Act of 1977. During these periods, the DOE established large-scale nuclear facilities to support both defense and civilian objectives. Defense-related goals included the design, production, and testing of nuclear weapons originating from the Manhattan Project. Civilian applications focused mainly on energy generation [12]. This widespread utilization of nuclear technology has resulted in the generation of a wide range of nuclear waste products. These waste products vary in composition and radioactivity, which requires complex strategies for their management and disposal to mitigate environmental and public health risks.

A significant turning point in nuclear waste production occurred with the Manhattan Project during World War II. The establishment of the Hanford Nuclear Site, shown in Figure 1.1 in Washington State exemplifies the scale of nuclear production efforts at the time. Hanford was integral in the production of 75 tons of plutonium, ^{239}Pu, for the atomic bombs used in Hiroshima and Nagasaki. The operations at Hanford, and similar facilities, resulted in the accumulation of substantial quantities of high-level waste (HLW), federal law, specifically 33 USCS 1402 (j) previously classified HLW as both spent fuel from a nuclear reactor and the by-products resulting from processing this fuel to recover other materials such as uranium and plutonium. However, this classification has since been redefined based on the level

of radioactivity in the waste, rather than how the waste was generated; therefore, HLW now refers to waste composed of hazardous fission products and transuranic elements with long half-lives [1]. The management of this waste presented an unprecedented challenge, as it required secure containment to prevent environmental contamination over extended periods.

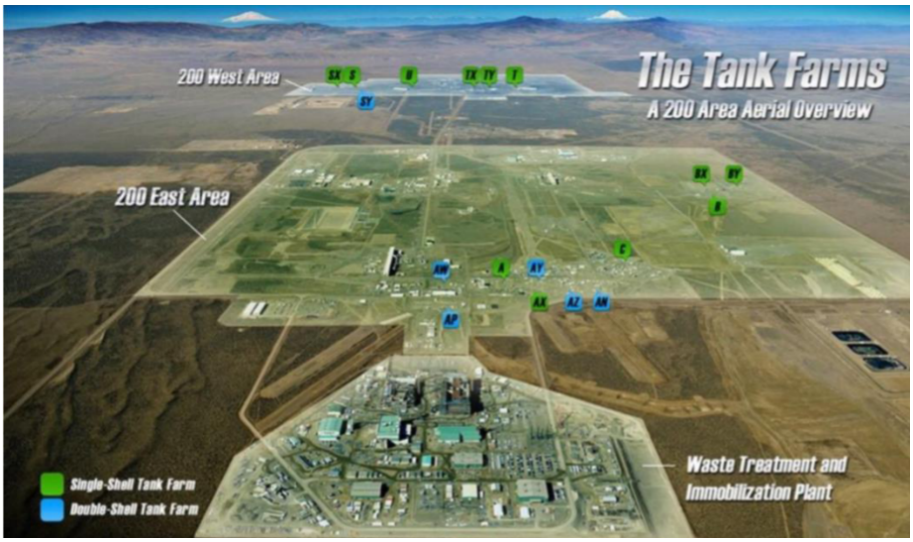

Figure 1.1: Aerial view of the Hanford Site 200 area.

The environmental impact of nuclear waste has been a growing concern, particularly with respect to the long-term stability of storage solutions. Early containment methods, such as the construction of 75 feet in diameter single-shell tanks (SST) shown in Figure 1.2 at sites such as Hanford, were initially intended as temporary measures. Established in 1944, Hanford now covers approximately 580 square miles of desert in southeastern Washington. The SSTs, constructed of concrete and carbon steel, were designed with a lifespan of 20 years to store high-level waste from plutonium production. However, leaks were detected in some SSTs, and over time, 67 SSTs have been confirmed or suspected to be leaking. This situation highlighted the limitations of early containment solutions and the release of radioactive con-

taminants into soil and groundwater, posing significant environmental and public health risks.

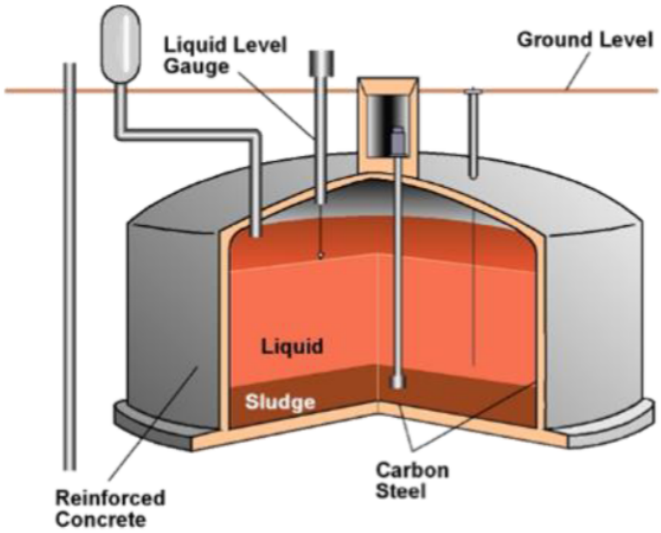

Figure 1.2: Schematic of a single-shell tank (SST) showing carbon-steel liner, reinforced-concrete dome, liquid and sludge layers, risers, and instrumentation.

In response to these failures, the Hanford site introduced larger, more secure double-shell tanks (DSTs) shown in Figure 1.3 in the 1960s, designed to last 50 years and feature a secondary shell for additional containment. Nearly all liquid waste, or supernate, has been relocated to these DSTs to ensure better public and environmental safety. Despite these advancements, the remaining sludge waste will eventually need to be stored in a permanent geological repository.

Figure 1.3: Visual of a DST under construction in 1975 [1].

The enduring challenge of managing nuclear waste is underscored by the long half-lives of radionuclides, which remain hazardous for thousands of years. To address these risks, extensive efforts are underway to develop long-term waste management solutions. One such approach is the vitrification of high-level waste, where radioactive waste is mixed with silica and melted into a stable glass. This method, currently being implemented at Hanford, aims to immobilize the waste and minimize its potential to leach into the environment. Nonetheless, challenges persist, including accurately characterizing legacy waste, retrieving waste from deteriorating storage tanks, and establishing permanent geological repositories for final disposal.

The production of nuclear waste and its resultant environmental impacts continue to be a focal point in discussions on the future of nuclear technology. As the legacy of nuclear production persists, the need for scientifically robust and sustainable waste management strategies remains critical to mitigating the environmental hazards posed by radioactive materials. The ongoing efforts to address these challenges are central to ensuring that nuclear waste's impact on the environment and public health is minimized.

1.2 Operational Context and Motivation

As the primary contractor for the Department of Energy Office of River Protection (DOE-ORP), engineers at the Hanford Site are responsible for the characterization, retrieval, treatment, and closure of underground tanks that contain radioactive and chemically contaminated liquid, sludge, and hardened residual waste. A major challenge within this mission is safely and effectively sampling materials in the high-risk environments inside these waste tanks. Traditionally, sampling is performed through a riser, which provides a vertical access point into the tank. The term "off-riser" refers to conducting operations outside the vertical reach of an open riser, allowing access to horizontal locations that cannot be reached with conventional tools due to limited riser placement. Traditional methods, such as deploying rovers, core samplers, or clamshell-style retrieval tools, have shown significant limitations in capability and ease of deployment. As a result, ongoing research efforts have focused on developing a more effective and versatile off-riser sampling solution.

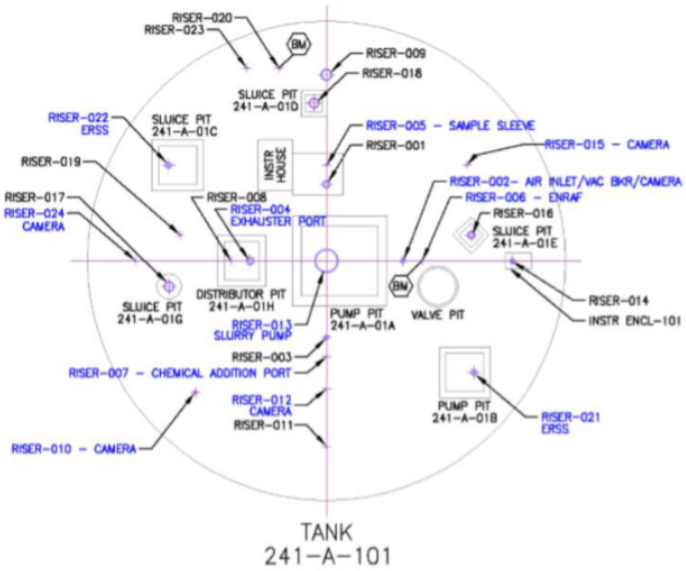

Figure 1.4: Riser layout and recommended utilization for Tank 241-A-101 [2].

Tank sampling and monitoring support several critical functions, including the characterization of tank waste for safe storage, transfer, and future delivery to the Waste Treatment Plant (WTP). Additionally, they aid in the assessment and mitigation of corrosion by providing the data necessary for modeling tank conditions, ensuring waste compatibility, and supporting SST waste retrieval operations. These activities help maintain tank structural integrity and ensure a safer work environment. Current methodologies for acquiring tank samples include core samples, grab samples, corrosion sampling, vapor sampling, vadose zone sampling, residual solids sampling, and liquid level monitoring.

1.2.1 Waste Tank Constraints and Technology Requirements

The design and implementation of any waste retrieval system for the Hanford Site's A and AX Tank Farms must adhere to several critical constraints to ensure operational efficacy and safety. These constraints are shaped by the tank environment, the physical properties of the waste, and the need to minimize additional risks during the waste retrieval process. Meeting these constraints is essential to avoid equipment failure, environmental contamination, or endangering personnel involved in the operation [3].

Riser Size and Limitations

The system must fit through a maximum 42-inch inner-diameter riser, which is the primary access point to the tank, or a smaller 4-inch or 12-inch riser for specific systems. These size constraints necessitate a compact yet functional system capable of passing through limited access points, as shown in Tank 241-A-101 in Figure 1.4, while maintaining retrieval effectiveness. Designing the system to fit through these

confined spaces while retaining operational capabilities is crucial for deployment in existing infrastructure without extensive tank modifications [3].

Tank Diameter Coverage

The system must be able to reach waste spread across the entire 75-foot diameter of the tank like the one shown in Figure 1.5, depending on the application of the technology. This requirement ensures that no areas within the tank are left unaddressed, maximizing waste retrieval efficiency. The system must also access the waste through multiple risers, allowing for flexible deployment to reach various parts of the tank, especially when facing obstructions or equipment in the tank [3].

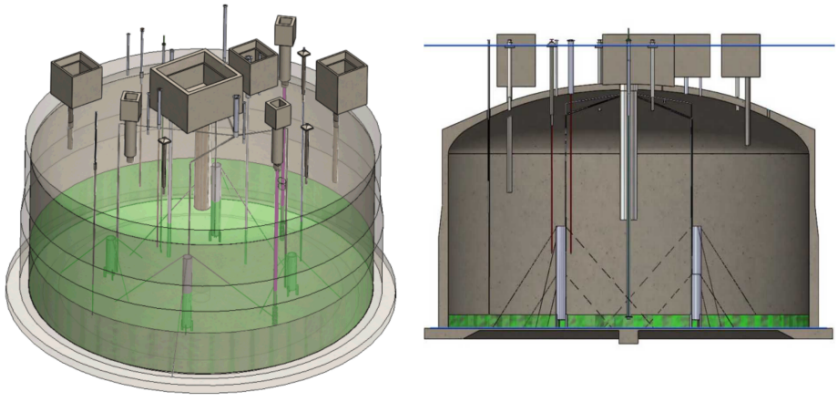

Figure 1.5: Three-dimensional model of an A-Farm waste tank: isometric (left) and cross-sectional (right) views with risers and internal hardware [3].

Depth and Movement Constraints

The retrieval system must be capable of reaching the bottom of the tank, approximately 60 feet below grade, and must be able to move or rotate to access waste around the tank's perimeter. Additionally, the system should be able to navigate around potential in-tank obstructions, such as air lift circulators, thermocouple trees, liquid level indicators, and uneven surfaces. The ability to maneuver through

these obstacles is essential to prevent the system from becoming stuck or damaged, which could halt the retrieval process and lead to costly delays [3].

Radiation Tolerance

The system must tolerate radiation levels of approximately 1,000 rad/hour gamma and up to 50,000 rad/hour beta. Figure 1.6 shows the historical survey data of gamma logging and spectral log via drywells. These radiation levels are hazardous to both humans and sensitive electronics, making radiation-hardening a critical design requirement for components exposed to the tank environment. Ensuring that the system can withstand such extreme conditions is essential for maintaining functionality during prolonged operations [3].

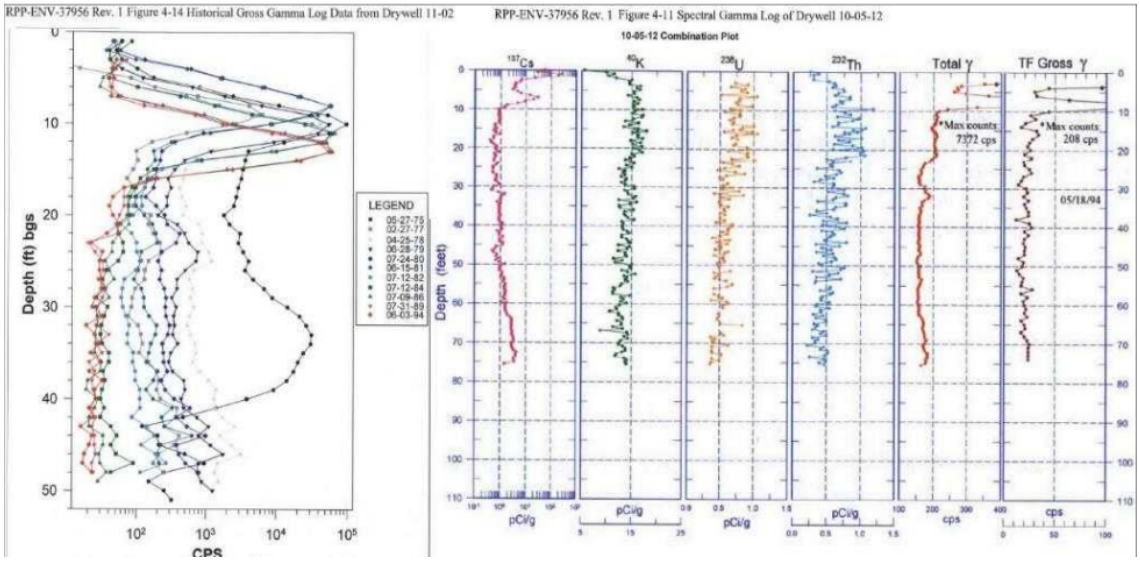

Figure 1.6: Historical radiological survey data: depth profiles and spectral gamma logs from prior site characterization [4].

Environmental Conditions

The system must be able to operate in temperatures ranging from 50 to 200 °F and in pH levels between 7 and 15. These conditions reflect the wide variability present

within tank environments, including exposure to highly corrosive chemical species. To address this issue, the system incorporates materials and components selected for chemical compatibility, thermal stability, and sustained mechanical performance under elevated temperatures. Furthermore, the system must operate reliably in environments with up to 100 percent relative humidity. This broad operational envelope ensures that the system remains functional across diverse tank conditions, preventing degradation or failure due to temperature fluctuations, corrosive constituents, or moisture infiltration [3].

Additionally, the technology must be spark proof due to the potential presence of hydrogen gas within the tank. This requirement is essential because certain electrical components or motor types can generate ignition sources during operation. To mitigate this risk, the system utilizes non sparking actuation methods and avoids electronics capable of producing arcs or sparks, thereby reducing the probability of ignition and supporting safe deployment in environments where combustible gases may be present.

Waste Management and Liquid Control

The retrieval process must carefully manage the amount of liquid used to mobilize and retrieve solid waste, as excessive liquid increases the volume of supernatant, complicating waste treatment and storage. The system should also minimize the amount of free-standing liquid remaining in the tank after retrieval. Excess liquid can hinder subsequent operations and complicate regulatory compliance for waste storage. Therefore, efficient liquid management is essential to maintaining safe and cost-effective waste retrieval operations [3].

Design Implications

These constraints directly inform the ORSS architecture and component choices. Riser access (down to 4 in) and the need to reach beyond the riser footprint drive a compact hanging deployment with an articulated arm rather than floor locomotion; 75 ft diameter coverage and internal obstructions motivate long-reach kinematics with a slim profile and modular joints. High radiation levels favor pneumatic actuation and radiation-tolerant, preferably passive or shieldable sensing, while the hydrogen ignition hazard requires spark-proof practice (no brushed motors, intrinsically safe wiring, and grounded enclosures). Finally, liquid-management limits steer the system toward predominantly dry mechanical end effectors (hammer/chisel, drill/auger, scoop) with optional vacuum/containment attachments to minimize added supernatant and prevent free-liquid carryover.

1.2.2 Initial Development

The initial research and development efforts that preceded this thesis laid the foundation for the work presented here. These efforts focused on addressing the critical need for innovative technologies and methodologies for waste retrieval and characterization at nuclear waste storage facilities, particularly at the Hanford Site. Over time, various stakeholders, including the Department of Energy Office of Environmental Management (DOE-EM) and Hanford contractors, have developed tools and systems designed to safely and efficiently sample, retrieve, and analyze nuclear waste. The progress achieved during these early stages has culminated in the current study, which aims to further refine and advance waste sampling technologies to support safer and more effective remediation efforts.

Commercial Technology Investigations

Robotic manipulators represent a crucial advancement in automation technology, offering versatile solutions for tasks ranging from industrial manufacturing to hazardous environment operations. These systems can be broadly classified on the basis of their kinematic structure, workspace geometry, and degrees of freedom. Understanding the characteristics and capabilities of different types of manipulators is essential for selecting and implementing the most appropriate design for specific applications.

Selective Compliance Assembly Robot Arm (SCARA) robots, which were initially developed at Yamanashi University in Japan in the early 1980s, have now become a standard in industries focused on precision assembly and material handling. An example can be found in Figure 2.3. Their distinctive design combines the benefits of cylindrical and articulated robot configurations, featuring selective compliance that is especially beneficial for vertical assembly operations.

The standard layout of an SCARA robot includes two parallel revolute joints that manage horizontal movement, a prismatic joint for vertical motion, and a final revolute joint for orienting the end effector. This configuration provides four degrees of freedom (4-DOF): translations in the X, Y, and Z axes, and rotation around the Z-axis (Rz). The parallel link structure inherently ensures high rigidity in the vertical direction while maintaining flexibility in the horizontal plane, which is the key feature and advantage of SCARA robots. Leading manufacturers such as Epson, ABB, Devonics, and Yamaha have developed extensive lines of SCARA robots with varying specifications. The SCARA design has proven particularly effective in clean room environments due to its ability to minimize particle generation and contamination. Modern SCARA robots often incorporate advanced features such as integrated vision systems, force sensing, and sophisticated control algorithms

that enhance their capabilities while maintaining their fundamental advantages in precision assembly applications [13].

An initial feasibility study explored the use of commercially available technology, specifically the Universal Robotics UR5e robotic arm shown in Figure 1.7, to function as an Off-Riser Sampling System (ORSS). The system appeared promising because of its radiation-hardened platform and the ability to be remotely controlled. Equipped with a depth camera, the UR5e demonstrated the potential for incorporating computer vision solutions, allowing for material sampling within the tanks. However, key issues arose, including the fact that the robotic arm could not fit within the four inch risers commonly used in Hanford tanks, making it impractical for locations with limited riser access. Additionally, external sensors such as the Intel RealSense D455 depth camera were sensitive to radiation, creating challenges for deployment in high radiation environments. Despite these limitations, the study successfully demonstrated that robotic arm based sampling was feasible. These findings motivated the investigation of alternative solutions better suited to the unique operational conditions of nuclear waste recovery [5].

Figure 1.7: UR5e semi-autonomous concept for off-riser sampling: vertical carriage with funnel and manipulator integration [5].

The success of SCARA robots in industrial applications caused the initial desire to integrate the technology for the Hanford site application discussed previously. In addition, it provides valuable information for specialized manipulator designs, such as the ORSS. Although the operating environment and requirements differ significantly, the principles of selective compliance and mechanical simplicity demonstrated by SCARA robots offer important considerations for manipulator design in nuclear waste handling applications. The ability of SCARA robots to achieve high rigidity in specific axes while remaining compliant with others can be a useful design philosophy when creating manipulators that need to perform precise tasks in confined or hazardous environments. Furthermore, the integration of advanced sensing and control systems in SCARA robots highlights the potential benefits of incorporating

similar technologies in the ORSS to improve safety, precision, and adaptability in the complex and highly regulated nuclear waste management domain.

Preliminary Iteration

An initial design concept based on a SCARA style robot, shown in Figure 1.8, was developed as an evolution of the UR5e concept. While most industrial robotic arms feature six axes for complete maneuverability within their designated workplace's, the initial assumption was that a four-axis configuration would suffice for sampling the area beneath the arm. If needed, future iterations could incorporate additional axes of motion to enhance functionality and tooling options [6].

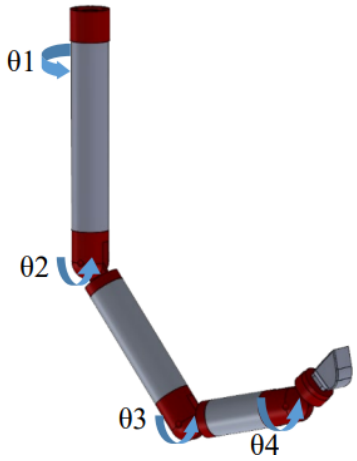

Figure 1.8: Kinematic concept of a 4-DoF articulated arm, indicating the four revolute joints and motion directions [6].

Building on this conceptual design, a proposed final solution concept, depicted in Figure 1.9, was developed to advance the system architecture. This concept centers on a hanging configuration that incorporates retractable stability elements to ensure secure positioning during sampling operations. The design includes the four degree of freedom arm described previously, which improves the system's ability to maneuver within confined spaces. This approach aims to address the unique challenges of

sampling inside the Hanford tanks while providing a robust and adaptable solution for waste sampling activities.

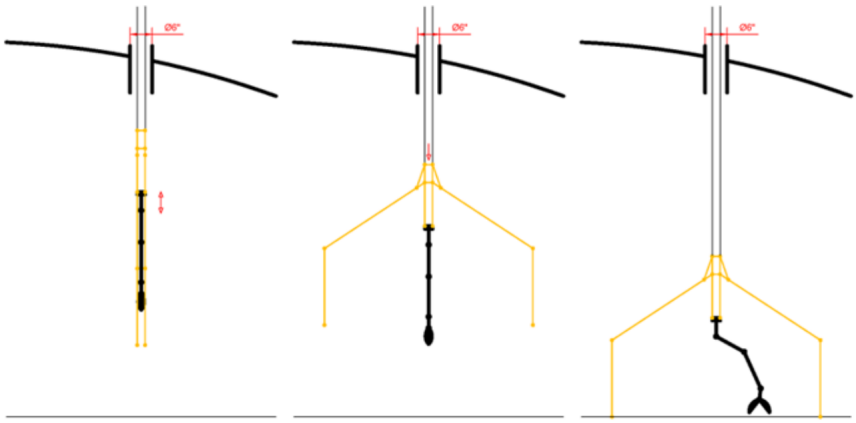

Figure 1.9: Early concept sketches for the final sampling system showing vertical probe, bifurcated support, and articulated manipulator variants [6].

The initial design of the joint, shown in Figure 1.10 introduced the concept of a modular system, opting for pneumatic actuation as a key design choice. The primary consideration in this phase was selecting an appropriate actuator to facilitate the joint movements. Three types of actuators were evaluated: electric, pneumatic, and hydraulic. Electric motors were deemed unsuitable due to their size and insufficient torque for handling the sludge waste, which resembles wet beach sand. In contrast, both hydraulic and pneumatic systems could meet the application's demands. Although hydraulics were acknowledged for their rigidity and ability to handle high loads, pneumatics were ultimately chosen for their ease of use. The conceptual design for the joints features a chain drive operated by a pneumatic cylinder, allowing for a larger cylinder to be integrated and concealed along the member's length. These components, known as joint members, offer enhanced force and a broader range of motion compared to other actuation methods. The payload capacity and motion range are influenced by the size of the air cylinder and the radius of the

sprocket. Should hydraulics be pursued in future iterations, the fluid used in the final design will be further assessed [6].

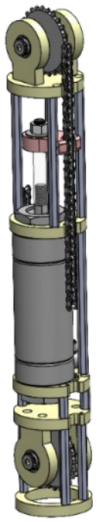

Figure 1.10: First design of the joint member with integrated pneumatic cylinder and chain–sprocket transmission [6].

On an industrial level, companies like ATIS pneumatic manipulators, has developed pneumatic manipulators similar what is found is Figure 2.4 to perform task like SCARA robots. Pneumatic manipulators are especially useful in environments where traditional electric or hydraulic robots are unsuitable due to fire hazards or other safety concerns like ones mentioned in Section 1.2.1. This advantage makes pneumatic systems particularly relevant for material handling in chemical industries. As discussed by Biswas Palock [14], a 3-axis articulated pneumatic robotic arm was developed and assembled to address such needs, demonstrating that pneumatic systems can offer a reliable and safer alternative for specific industrial applications. The integration of pneumatic actuators provides compliance and safety benefits in environments where electrical systems pose significant risks, thereby expanding the versatility of robotic manipulators in sensitive and hazardous conditions.

Second Iteration

The redesign of the previous iteration, shown in Figure 1.11, focused on enhancing modularity, resulting in a joint core design that consists of interchangeable components like the driving end, motion carriage, idler end, and actuator core. This modularity allows for easier updates and modifications in the future. The design's strength comes from using aluminum standoffs and steel linear rails, while some parts are 3D printed with ABS plastic, with future plans to machine critical components from aluminum for durability [6].

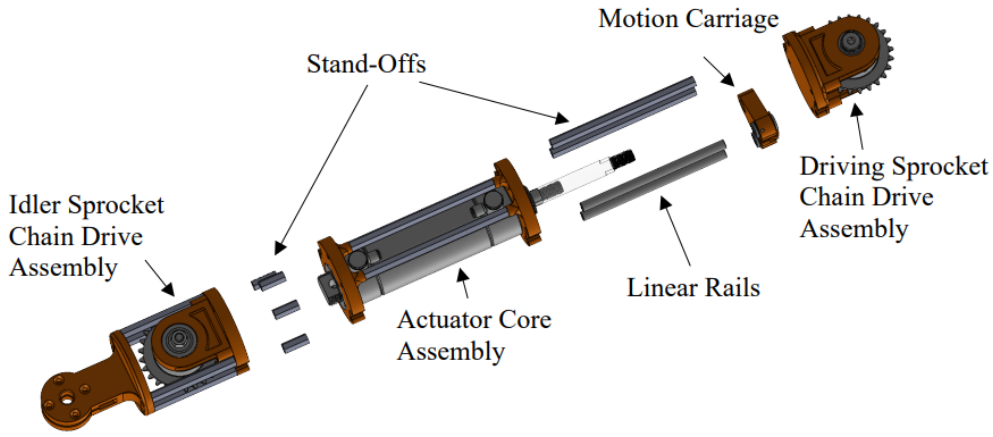

Figure 1.11: Exploded view of the second-iteration actuator module showing driving and idler sprocket chain drives, motion carriage, linear rails, stand-offs, and actuator core assembly [6].

The drive system uses a roller-chain sprocket connected to the next joint, supported by a linear motion carriage that transfers movement from an air cylinder. The carriage also includes a tensioner to maintain smooth operation. The air cylinder itself doubles as a structural component to reduce weight, and smaller actuators are used closer to the end effector to handle reduced loads, further decreasing overall weight [6]. A three degree of freedom prototype can be shown in Figure 1.13.

The arm's reach is approximately 4 feet, shown in Figure 1.12, and the end effector will include a tool changer with multiple tools for handling various waste types [6]. These tools will be customized to address different consistencies of waste, including sludge, granular materials, and liquid residues, ensuring versatility across multiple waste retrieval scenarios. The tool changer will enhance the system's adaptability, allowing operators to quickly switch between tools depending on the task at hand, ultimately improving the efficiency and effectiveness of the waste sampling and retrieval process [8].

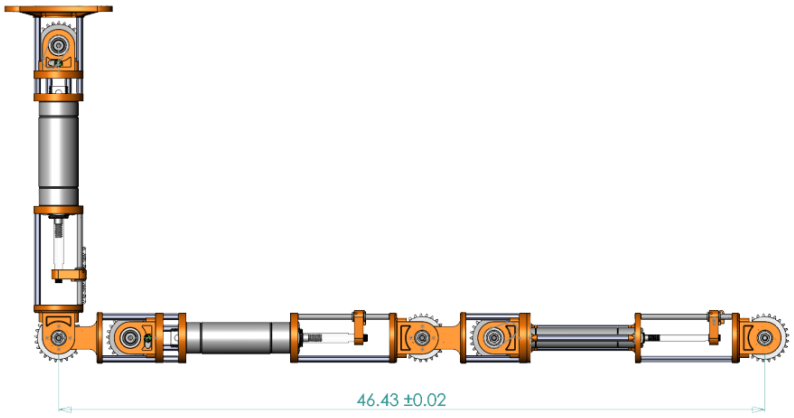

Figure 1.12: Second-iteration manipulator in fully actuated, extended configuration with overall reach dimension [6].

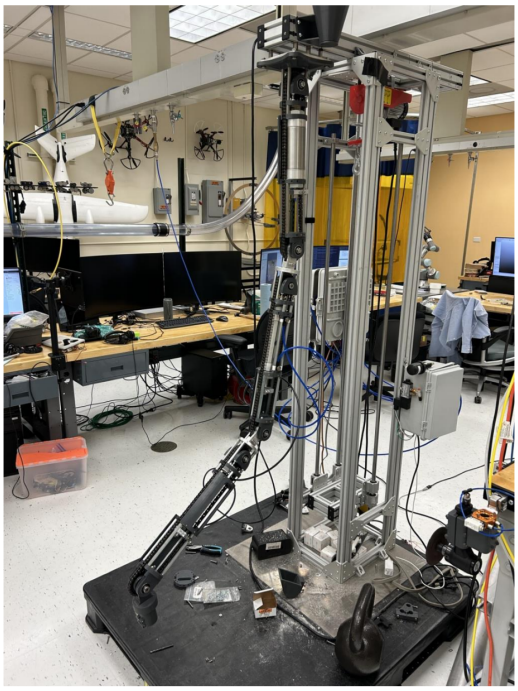

Figure 1.13: Second-iteration prototype with three degrees of freedom (3-DoF) assembled for laboratory testing [7].

Tertiary Iteration

The third design, shown in Figure 1.14, maintains the same fundamental mechanism as the previous iteration, converting the input force of the pneumatic actuator into the output moment at the axis of rotation. However, this design introduces some material changes and additional components. The main body is now constructed from aluminum, specifically AL6061-T6, chosen for its lightweight properties and cost-effectiveness. The chain clamp has been fabricated from mild carbon steel to ensure enhanced strength [8]. Key components of this design include:

- Stainless steel pneumatic actuators with varying bore sizes (1.5 inches to 3.5 inches), tailored to meet the specific torque requirements of each joint.

- A custom carbon steel adapter (shown in Figure 1.15) that connects the linear motion of the pneumatic actuator to an ANSI roller chain, facilitating the rotational motion of the elbow joint.
- The primary structure made from custom AL6061-T6 aluminum components.
- Carbon steel components including the chain, turnbuckle, clevis, and sprocket.
- 304 stainless steel bearings, pneumatic fittings, and hardware.
- Custom bar extensions connecting different assembles together instead of the hexagonal stand-offs, addressing torsion issues experienced in previous design
- A new addition of an in-line chain tensioner, a clevis turnbuckle design, to improve chain performance and longevity.

This design retains the efficiency of the previous iteration while incorporating material changes for improved performance and the addition of the in-line chain tensioner for enhanced functionality.

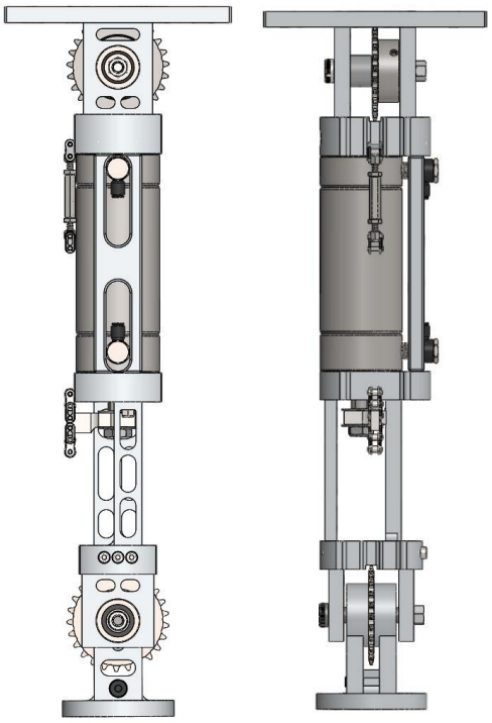

Figure 1.14: Joint assembly, Iteration 3: front and side views highlighting actuator integration and transmission layout [8].

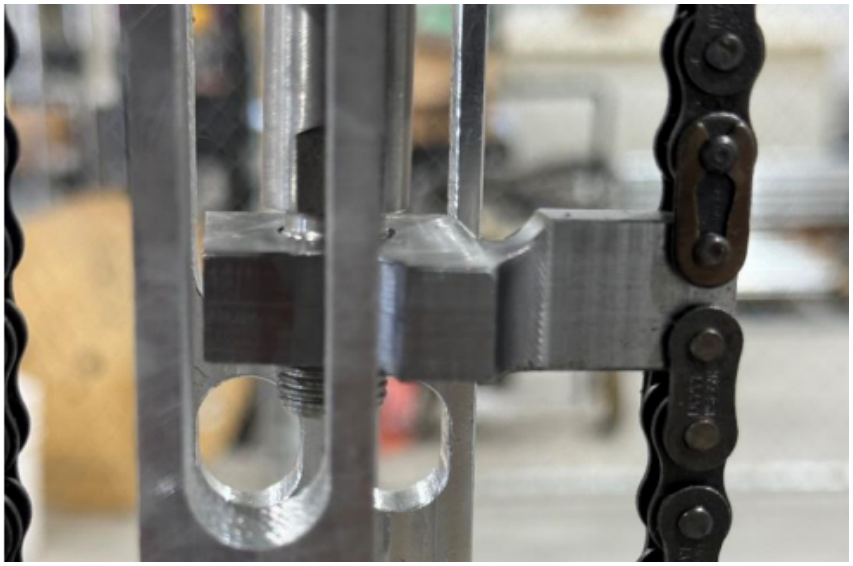

Figure 1.15: Carbon-steel chain clamp used to secure the drive chain to the carriage [8].

Hanford On-Site Demonstrations

Building upon the description of the third design iteration, it is important to note that while this version introduced significant improvements, certain limitations became apparent during testing and evaluation on site. Following consultations with operators and members of the Sampling Group, who oversee sampling procedures and strategies for SSTs and DSTs at the Hanford site, a comprehensive list of desirable end effectors was developed. All items were either 3D printed using PETG black filament or sourced commercially.

The primary objective of this effort was to demonstrate the system's performance and behavior with various end effectors, thereby validating the concept and illustrating its potential. The desired end effectors included: an Auger, shown in Figure 1.16, designed to sample porous materials such as sand; Drills, shown in Figure 1.17, intended to puncture materials and serve as a preparation step for the auger; a Chisel, shown in Figure 1.18, developed to fracture material through impact created by bit oscillation; and a Scoop, shown in Figure 1.19 [8].

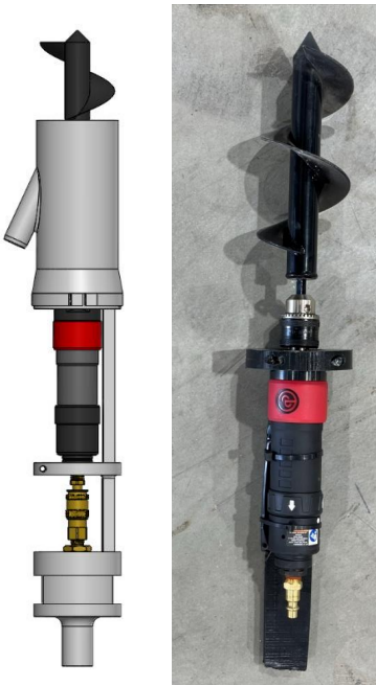

Figure 1.16: Pneumatic drill end effector with 3-inch auger bit and integrated vacuum attachment port: CAD section (left) and fabricated unit (right) [8].

Figure 1.17: Pneumatic drill end effector with 3/8-inch (0.375 in) carbide bit: CAD section (left) and fabricated unit (right) [8].

Figure 1.18: Pneumatic hammer end effector with pointed chisel bit: CAD section (left) and fabricated unit (right) [8].

Figure 1.19: Prototype 3D-printed scoop integrated with the manipulator linkage [8].

The system still faced challenges with stability during operation, particularly when extended to its full reach. This instability could potentially compromise the accuracy of sampling and the safety of operations in the confined and hazardous environment of nuclear waste tanks. Furthermore, the current design, while versatile, was limited in its ability to handle the wide variety of waste consistencies present

in the tanks. The end effector options were not sufficiently diverse to address all potential sampling scenarios efficiently [8].

This thesis builds upon research at Florida International University's Applied Research Center (FIU-ARC), conducted in collaboration with the Department of Energy and Hanford Site. The work focuses on advancing waste sampling technologies for nuclear waste storage facilities, particularly at the Hanford Site.The research progressed through several iterations, starting with investigations into commercial technologies like the Universal Robotics UR5e robotic arm. While proving the concept of robotic sampling, this approach revealed limitations in fitting through small risers and radiation sensitivity of external sensors. These challenges, along with the unique constraints of nuclear waste environments, motivated the redesign and development of a custom system.

Subsequent iterations refined the design, moving towards a custom four-degree-of-freedom arm with pneumatic actuation. The second iteration introduced a modular joint core design, while the third iteration further improved performance with material changes and additional components. Throughout this process, the research has been driven by the need to address conceptual and design issues specific to nuclear waste sampling, including size constraints, radiation tolerance, and the ability to handle various waste consistencies. This thesis focuses on the latest design iteration, exploring in detail its capabilities and potential impact on improving the safety and efficiency of nuclear waste retrieval operations. Additionally, it addresses the development of control systems necessary for precise and reliable operation in hazardous environments. By tackling these conceptual and design challenges, this research aims to significantly advance the field of nuclear waste management and remediation.

CHAPTER 2

LITERATURE REVIEW

Building upon the comprehensive development and testing of the Off-Riser Sampling System (ORSS), this chapter presents a thorough literature review to contextualize our work within the broader field of robotic manipulation and control, particularly in hazardous environments. The review aims to provide a solid theoretical foundation for the advancements proposed in our system, including improved control mechanisms and positions ORSS against current tank-sampling technologies that motivate our design choices.The literature review will cover several key areas crucial to the development and optimization of the ORSS:

- Related Work: Current Tank Sampling Technologies: A focused comparison of fielded sampling systems (e.g., ORSS crawler, auger samplers, clamshell/drag/core tools), highlighting riser-access constraints, hard-heel handling, mobility, and sample quality; this establishes the concrete gaps our approach targets.
- Robotic Manipulators: We will examine various forms of manipulators, focusing on those designed for operation in challenging and hazardous environments. This section will explore different kinematic structures, materials, and design considerations that inform our approach to the ORSS.
- Mathematical Models of Manipulators: To better understand and predict the behavior of our system, we will dive into mathematical modeling of robotic arms. This will include kinematic and dynamic models, which are essential for precise control and stability enhancement.
- Control of Pneumatic Actuators: Given the ORSS's reliance on pneumatic actuation, we will review state-of-the-art control strategies for pneumatic sys-

tems. This section will cover both linear and nonlinear control methods, addressing challenges such as compressibility and friction that are inherent in pneumatic systems.

By thorough examination of these areas, this review of the literature will provide the theoretical background necessary to support and guide the ongoing development of the ORSS. It will help identify best practices, potential challenges, and innovative solutions that can be applied to our system, ensuring that our future work is well grounded in current scientific understanding and engineering principles.

2.1 Related Work: Current Tank Sampling Technologies

Off Riser Sampler System

The Hanford Off-Riser Sampling System (HORSS), shown in Figure 2.1, is a remote-operated mobile sampler consisting of a sampling device and a carrier. It is designed to collect residual waste samples from the floor of storage tanks after waste retrieval operations. The system is deployed into a tank through a riser with a diameter of 12 inches or larger. Once inside, the HORSS maneuvers remotely to gather samples of the remaining waste. The collected material is then placed into a jar housed within the carrier and subsequently raised into a glove bag located at the top of the riser. This process is repeated until the required number of samples has been collected [4].

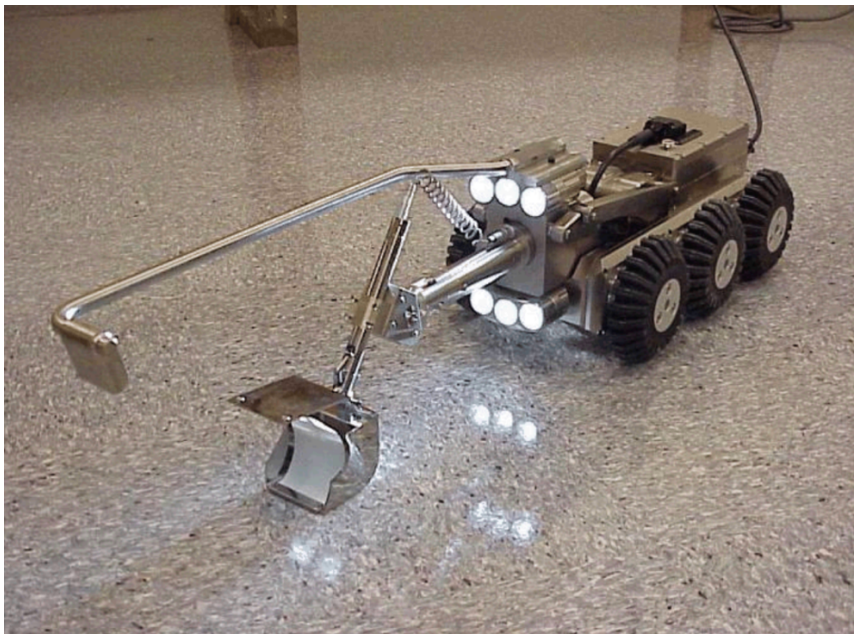

Figure 2.1: General Electric wheeled inspection robot for off-riser tank inspection, with onboard lighting and sampling/manipulator arm [4].

Sampling of residual tank waste is a crucial step in the tank retrieval and closure process. However, current sampling methods, including the ORSS, face limitations in reliably collecting hard heel material not directly beneath the riser. The existing ORSS, a General Electric Inspection Technology crawler equipped with a clamshell sample scoop, is not robust enough to break off pieces of hard heel for sampling without additional mechanical support from an attached sample crusher (often referred to as a tenderizer) [3].Further challenges associated with the ORSS include a minimum required riser diameter of 12 inches, which limits its use in certain tanks. Additionally, the sample scoop's limited strength affects its ability to gather more solid materials effectively. The thin wheels of the unit can cause it to sink into soft or sandy materials, and its low clearance makes it prone to getting caught on waste material or debris within the tank [4].

Auger Sampling System

The auger sampling system, shown in Figure 2.2, is designed for deployment through risers 4 inches in diameter or larger, utilizing a motorized drill unit similar to a demolition hammer but compacted to fit a 3.5-inch form factor. It uses a 1.5-inch masonry drill bit connected to a sample collection vessel that transports waste samples to a collection cup above the tank. The system can retrieve up to 200 ml of non-stratified samples by penetrating approximately 7 inches into the waste. Modular and adaptable, the system includes a retractable drill bit and torque reaction teeth for stability. The unit is contained in a glove bag for contamination control, and it disassembles into a small waste package for disposal, with reusable components [4].

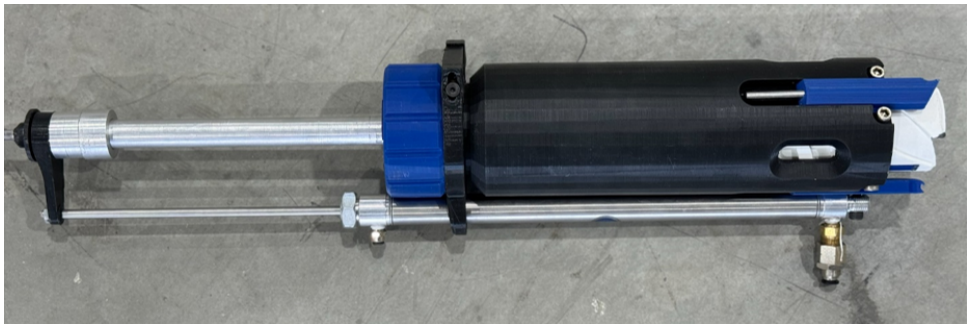

Figure 2.2: Prototype auger-based sampling tool showing drive rod, guide rails, and housing [4].

Summary of Gaps. Current fielded tools expose four practical gaps that drive the design: (i) riser access constraints ($\geq$ 12 in) limit deployment to large openings, motivating an off-riser reach architecture; (ii) clamshell scoops cannot reliably fracture hard-heel deposits, necessitating percussive and penetrative tooling (chisel/hammer and drill/auger); (iii) thin, low-clearance wheelsets tend to bog or snag in heterogeneous sediments, favoring a hanging/anchored mechanism over floor locomotion; and (iv) several methods bias samples toward surface layers, requiring

depth-capable end effectors to target subsurface material. The proposed ORSS addresses these in turn via a compact, suspended manipulator that reaches beyond the riser footprint; a modular end-effector suite including pneumatic hammer/chisel and drill/auger for hard-heel breakup and recovery; a stabilized, non-wheeled support that eliminates traction failure and entanglement risks; and tooling and procedures designed for controlled penetration and retrieval at depth, improving representativeness of collected samples.

2.2 Robotic Manipulators

Robotic manipulators represent a crucial advancement in automation technology, offering versatile solutions for tasks ranging from industrial manufacturing to hazardous environment operations. These systems can be broadly classified on the basis of their kinematic structure, workspace geometry, and degrees of freedom. Understanding the characteristics and capabilities of different types of manipulators is essential for selecting and implementing the most appropriate design for specific applications.

2.2.1 SCARA Robots

Selective Compliance Assembly Robot Arm (SCARA) robots, which were initially developed at Yamanashi University in Japan in the early 1980s, have now become a standard in industries focused on precision assembly and material handling. An example can be found in Figure 2.3. Their distinctive design combines the benefits of cylindrical and articulated robot configurations, featuring selective compliance that is especially beneficial for vertical assembly operations.

The standard layout of an SCARA robot includes two parallel revolute joints that manage horizontal movement, a prismatic joint for vertical motion, and a final revolute joint for orienting the end effector. This configuration provides four degrees of freedom (4-DOF): translations in the X, Y, and Z axes, and rotation around the Z-axis (Rz). The parallel link structure inherently ensures high rigidity in the vertical direction while maintaining flexibility in the horizontal plane, which is the key feature and advantage of SCARA robots. Leading manufacturers such as Epson, ABB, Devonics, and Yamaha have developed extensive lines of SCARA robots with varying specifications. The SCARA design has proven particularly effective in clean room environments due to its ability to minimize particle generation and contamination. Modern SCARA robots often incorporate advanced features such as integrated vision systems, force sensing, and sophisticated control algorithms that enhance their capabilities while maintaining their fundamental advantages in precision assembly applications [13].

The success of SCARA robots in industrial applications caused the initial desire to integrate the technology for the Hanford site application shown in Section 1.2.2. In addition, it provides valuable information for specialized manipulator designs, such as the ORSS. Although the operating environment and requirements differ significantly, the principles of selective compliance and mechanical simplicity demonstrated by SCARA robots offer important considerations for manipulator design in nuclear waste handling applications. The ability of SCARA robots to achieve high rigidity in specific axes while remaining compliant with others can be a useful design philosophy when creating manipulators that need to perform precise tasks in confined or hazardous environments. Furthermore, the integration of advanced sensing and control systems in SCARA robots highlights the potential benefits of incorporating similar technologies in the ORSS to improve safety, precision, and

adaptability in the complex and highly regulated nuclear waste management domain.

Figure 2.3: Devonics Fairino FR20 collaborative robotic arm (side view) [9].

2.2.2 Pneumatic Manipulators

Pneumatic robotic manipulators constitute a distinctive category of robotic systems that employ compressed air for their actuation mechanisms. These manipulators are especially beneficial in settings where electrical components might be hazardous or where durable and economical solutions are necessary. Their basic functioning depends on compressed air to power linear or rotary actuators, which transforms pneumatic energy into mechanical motion to perform manipulation tasks.

On an industrial level, companies such as ATIS Pneumatic Manipulators have developed pneumatic systems similar to the example shown in Figure 2.4 to perform tasks typically associated with SCARA style robots. Pneumatic manipulators are especially useful in environments where traditional electric or hydraulic robots are unsuitable due to fire hazards or other safety concerns, such as those described in Section 1.2.1. This advantage makes pneumatic systems particularly relevant for

material handling in chemical and other safety critical industries. As discussed by Biswas Palock [14], a 3-axis articulated pneumatic robotic arm was developed and assembled to address such needs, demonstrating that pneumatic systems can offer a reliable and safer alternative for specific industrial applications. The integration of pneumatic actuators provides compliance and safety benefits in environments where electrical systems pose significant risks, thereby expanding the versatility of robotic manipulators in sensitive and hazardous conditions.

Figure 2.4: Commercial ATISMIRUS-80 counterbalanced manipulator on pedestal mount [10].

2.3 Mathematical Models of Manipulators

Mathematical modeling underpins analysis and control of robotic manipulators, including SCARA architectures and pneumatic-actuated arms. Kinematically, most studies adopt the Denavit–Hartenberg (DH) convention to parameterize open-chain geometry via per-link frames and the (α, a, θ, d) quadruple, yielding compact forward-

kinematics maps and Jacobians for velocity/force transmission analysis. DH tables also serve as the backbone for inverse-kinematics formulations and singularity assessment, and remain standard in recent SCARA-focused publications [15, 16].

On the dynamics side, two complementary formulations are widely used: the Euler–Lagrange energy based method and the recursive Newton–Euler force and moment balance algorithm. Both yield the canonical manipulator model $M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = \tau$, which supports inverse dynamics feedforward, computed torque control, and model based observers. Lagrangian derivations are often chosen for their symbolic structure and clear parameter grouping, whereas Newton–Euler recursions are preferred for their computational efficiency in real time control. Contemporary SCARA studies frequently present both approaches to cross validate dynamic coefficients prior to controller design [15, 16].

For SCARA robots (typically $R-R-P$ or $R-R-R-P$ layouts), the planar shoulder–elbow structure simplifies closed-form kinematics and exposes characteristic singular loci in the Jacobian. Recent works report DH-based kinematics, velocity/acceleration propagation, and full Lagrange/Newton–Euler dynamics, followed by identification or sensitivity analysis to refine link inertias and friction terms for accurate trajectory tracking [15, 16]. These models directly inform controller synthesis (PID with feedforward, computed-torque, and hybrid schemes) and performance metrics such as settling time and overshoot.

Pneumatically driven manipulators introduce additional states and nonlinearities from compressible flow, valve orifices, and chamber thermodynamics. Here, rigid-link Lagrangian or Newton–Euler dynamics are coupled with pneumatic submodels (mass-flow $\dot{m}(p,u)$, chamber pressure dynamics, effective compliance), producing grey-box structures whose parameters are identified experimentally. Studies on pneumatic arms and muscle-actuated joints show that incorporating pressure dy-

namics and load-dependent compliance markedly improves prediction fidelity and controller robustness, especially near contact and under varying loads [17]. In practice, these hybrid models—DH kinematics plus rigid-body dynamics augmented with pneumatic state equations—provide the foundation for gain scheduling, disturbance compensation, and hybrid bang–bang/flow-throttling controllers used in SCARA-like assemblies with pneumatic actuation.

2.4 Control of Pneumatic Actuators

Pneumatic actuators (linear cylinders and rotary vane motors) are attractive for their power density and robustness, but compressible fluid dynamics, valve nonlinearities, and friction make precise motion control challenging. Historically, simple on/off (bang–bang) strategies were used because they map directly to solenoid valve hardware, yet they tend to induce limit cycles near the setpoint. Contemporary work therefore emphasizes continuous controllers—chiefly PID—augmented by plant-aware compensation or switching logic to recover both accuracy and speed [18, 19, 20].

A widely cited study by Wang et al. developed a practical PID-based strategy for servo-pneumatic actuators, showing that careful gain selection, friction/feedforward compensation, and pressure feedback can overcome much of the nonlinearity and deliver robust tracking with modest computational cost [18]. In parallel, foundational modeling by Richer and Hurmuzlu formalized pneumatic actuator dynamics—valve flow, chamber pressure evolution, and line dynamics—providing the mass–flow and pressure states needed to interpret PID behavior (e.g., integrator windup and apparent delay) and to design stabilizing filters and anti-windup schemes [19]. Together,

these works frame PID as a viable baseline when paired with physics-informed tuning and simple compensators.

On/off control remains compelling for its speed and simplicity, especially with fast solenoids. Qi et al. analyze position control of double-acting cylinders using on/off valves under three-mode switching (e.g., open/close/PWM), demonstrating that high-frequency PWM can emulate proportional flow while discrete modes handle large errors efficiently [20]. The literature increasingly favors hybrid schemes: use bang–bang (or full-duty PWM) when the error is large to maximize differential pressure and transit quickly, then transition to a low-duty PWM/linearized regime or a PID inner loop near the target to suppress overshoot and chatter [20]. This division of labor preserves fast approach while achieving fine positioning.

Model-informed PID (with friction/feedforward terms and anti-windup) provides a tunable core suited to embedded controllers, while mode scheduling with on/off or PWM valves supplies the authority needed during large excursions [18, 20]. The high-fidelity pneumatic model in [19] explains observed transients (e.g., pressure build-up, dead zones) and guides selection of sampling rates, valve timing, and filters so that hybrid PID + switching controllers remain stable and repeatable across load changes. Overall, recent studies support PID as the priority method for precision, with bang–bang/PWM used strategically to extend speed and range.

CHAPTER 3

SYSTEM MATHEMATICAL MODEL

Achieving precise position control of a pneumatic actuator is critical to the effective operation of the ORSS. This control requires leveraging pressure, mass flow rate, and position feedback to ensure that the actuator responds accurately to system demands. To address this challenge, a mathematical model of the actuator is developed, capturing its dynamic behavior. This model is validated and simulated using a Simulink-based framework, enabling a detailed analysis of its performance under varying conditions. Furthermore, the actuator's position data is used to derive a kinematic model, integrating the actuator's behavior with the overall mechanical system. By developing and validating this control framework, the chapter lays the foundation for optimizing ORSS functionality, ensuring reliability and precision in operation.

3.1 Pneumatic Actuator Mathematical Model

The development of a mathematical model for a pneumatic actuator is essential for understanding and improving its dynamic behavior, particularly in applications requiring precise position control. This model aims to capture the relationships between pressure, mass flow rate, and actuator motion, offering insights into how these factors influence system performance. By the end, the model will provide a foundation for predicting actuator behavior, optimizing control strategies, and integrating the actuator into larger mechanical systems.

3.1.1 Mass Transfer Through a Pipe

Pressure drop (ΔP) in air transfer systems is caused by friction, turbulence, and pipe characteristics, leading to a reduction in pressure as air flows. This must be considered since the system will be operated remotely from a designated safe distance. The Darcy-Weisbach equation is commonly used to calculate pressure drop, accounting for factors like pipe length (L), diameter (d_p), and roughness (ϵ). While typically applied to incompressible fluids, modifications are necessary when using it for compressible fluids like air, considering factors such as changes in density (ρ) and velocity(v_p). Accurate modeling using this equation is crucial for efficient system design, minimizing energy losses, and preventing system inefficiencies and equipment wear.

$$\Delta P = P_{outlet} - P_{inlet} \tag{3.1}$$

$$P_{outlet} = f\frac{L}{d_p}\frac{\rho v^2}{2} + P_{inlet} \tag{3.2}$$

The pressure at the inlet of the pipe is denoted as P_{inlet}, while P_{outlet} represents the pressure at the outlet. For the specific case of air traveling toward the cylinder's chamber, the outlet pressure is equal to the supply pressure, i.e., $P_{\text{outlet}} = P_{\text{supply}}$. The velocity of air through the pipe is given by:

$$v = \frac{\dot{m}_i}{\rho\,\pi\left(\frac{d_p}{2}\right)^2} \tag{3.3}$$

where $\dot{m}_i$ is the mass flow rate, ρ is the air density, and d_p is the pipe diameter. Using the ideal gas law, the air density at the pipe inlet can be expressed as:

$$\rho = \frac{P_{\text{inlet}}}{RT} \tag{3.4}$$

Here, R is the specific gas constant, and T is the absolute temperature of the air.

The Darcy friction factor (f) quantifies the frictional resistance to flow within the pipe. It is influenced by the flow regime (laminar or turbulent), the Reynolds

number (Re), and the relative roughness of the pipe, defined as $\frac{\epsilon}{d_p}$, where ϵ represents the pipe's roughness height.

$$Re = \frac{\rho v d_p}{\mu} \tag{3.5}$$

where the dynamic viscosity (μ) is a function of temperature (T) and is represented by Sutherland's Law:

$$\mu(T) = \mu_0 (\frac{T_0 + S}{T + S})(\frac{T}{T_0})^{\frac{3}{2}} \tag{3.6}$$

The reference dynamic viscosity at a reference temperature T_0 is given by $\mu_0 = 1.716 \times 10^{-5}\,\mathrm{Pa \cdot s}$ at $T_0 = 273.15\,\mathrm{K}$. The Sutherland constant (S) for air is approximately $120\,\mathrm{K}$. The friction factor is a piecewise function, determined based on the flow regime:

- A Reynolds number less than 2000 indicates laminar flow.
- A Reynolds number greater than 4000 indicates turbulent flow.

For Reynolds numbers between 2000 and 4000, the flow is considered transitional, exhibiting characteristics of both laminar and turbulent flow. Swamee-Jain equation is used to empirically estimate the friction factor for fully developed turbulent flow in a circular pipe.

$$f = \begin{cases} \frac{64}{Re} & \text{if } Re < 2000, \\ \frac{0.25}{(\log(\frac{\epsilon/d_p}{3.7} + \frac{5.74}{Re^{(0.9)}}))^2} & \text{if } Re > 4000. \end{cases} \tag{3.7}$$

The pipe roughness, denoted by ϵ, represents the height of the irregularities or roughness on the interior surface of a pipe. The value of ϵ is standardized based on the material of the pipe and can be selected accordingly for the chosen pipe.

3.1.2 Mass Transfer Across an Orifice

The mass flow rate ($\dot{m}$) through an orifice is a critical parameter in pneumatic systems, representing the amount of fluid passing through per unit time. It is influenced by upstream (P_u) and downstream (P_d) pressures, the orifice area (A_o), the discharge coefficient (C_d), and fluid properties, requiring precise modeling for accurate system control [21], and is represented by

$$\dot{m}_i = A_0 C_d \Psi(P_d, P_u) \tag{3.8}$$

For airflow into a pneumatic cylinder, P_u is the pressure in the supply line (P_{supply}), and the downstream pressure is the pressure inside the cylinder ($P_{chamber}$). As air flows into the cylinder, known as charging the cylinder, P_u is higher than P_d, driving the flow until equilibrium is reached[21].

$$\text{Charging (flow into the cylinder):} \quad P_d = P_{supply}, P_u = P_{chamber} \tag{3.9}$$

$$\text{Discharging (flow out of the cylinder):} \quad P_d = P_{chamber}, P_u = P_{atomosphere} \tag{3.10}$$

The function $\Psi(P_d, P_u)$ is defined in a piecewise, non-linear manner, with its behavior dependent on the pressure ratio, which determines if the flow is choked or un-choked [21]:

$$P_{cr} = \frac{P_d}{P_u} = (\frac{2}{\gamma + 1})^{\frac{\gamma}{\gamma - 1}} \tag{3.11}$$

For air, the specific heat ratio (γ) is 1.4, which results in a critical pressure ratio (P_{cr}) of approximately 0.528. This means that if the downstream pressure falls below 52.8% of the upstream pressure, the flow becomes choked. In this condition, the mass flow rate reaches its maximum limit, and further reductions in downstream pressure will not increase the flow rate [21].

$$\Psi(P_d, P_u) = \begin{cases} \sqrt{\frac{2\gamma}{R(\gamma-1)}} \frac{P_u}{\sqrt{T_u}} (\frac{P_d}{P_u})^{\frac{1}{\gamma}} \sqrt{1 - (\frac{P_d}{P_u})^{\frac{\gamma-1}{\gamma}}} & \text{if} \quad \frac{P_d}{P_u} > P_{cr}, \\ \sqrt{\frac{\gamma}{R(\frac{\gamma+1}{2})^{\frac{\gamma+1}{\gamma-1}}}} \frac{P_u}{\sqrt{T_u}} & \text{if} \quad \frac{P_d}{P_u} \leq P_{cr}. \end{cases} \tag{3.12}$$

For air, $\gamma = 1.4$ and the universal gas constant, $R = 287 \frac{J}{KgK}$.

Flow is unchoked when $\frac{P_d}{P_u} > P_{cr}$, and it becomes choked when $\frac{P_d}{P_u} \leq P_{cr}$. Additionally, T_u denotes the upstream temperature and T_d denotes the downstream temperature. These temperatures can be determined using the ideal gas law together with the instantaneous total mass and pressure in the cylinder.[21].

$$T = \frac{PA_c x_{rel}}{mR} \tag{3.13}$$

Here, A_c denotes the cross sectional area of the chamber, and x_{rel} represents the position of the piston relative to the end of the charged region of the cylinder, including the dead space inherent to the chamber.

3.1.3 Pressure Variation in a Dynamic Cylinder

This subsection derives the pressure dynamics in the cylinder chambers by relating mass flow through the valve to pressure changes. Assuming adiabatic conditions, which are valid for fast-acting systems, and isothermal conditions, which are valid for slow-acting systems, the ideal gas law and thermodynamic principles are applied to establish the governing equation.

Adiabatic:

$$\dot{P}_i = \frac{\gamma R T \dot{m}_i}{x_{rel} A_c} - \frac{P_{chamber} \dot{x}}{x_{rel}} (\frac{\gamma R}{c_p} + 1) \tag{3.14}$$

Isothermal:

$$\dot{P}_i = \frac{R T \dot{m}_i}{x_{rel} A_c} - \frac{P_{chamber} \dot{x}}{x_{rel}} \tag{3.15}$$

where c_p is the specific heat capacity of air at a given temperature and $P_{chamber}$ is the instantaneous pressure of the chamber. The change in pressure will be calculated in both chambers of the cylinder, cap-side and rod-side, and will consider the m_i for both charging and discharging.

3.1.4 Force Balance in Cylinder

The force balance within the cylinder to determine the force exerted by the actuator, given the pressures in each chamber. The analysis accounts for key factors influencing the force output, including potential losses. Among these, friction plays a significant role, with its impact largely dictated by the cylinder's internal design and lubrication conditions. This force balance forms the foundation for accurately modeling the actuator's performance. Using Newton's second law of motion, $m_p\ddot{x} = F_{net}$, where m_p is the mass of piston and rod assembly, $\ddot{x}$ is the acceleration of the piston, and F_{net} is the net forces acting on the piston.

$$F_{net} = \delta(P_{cap}A_{cap} - P_{rod}A_{rod}) - F_{friction} - F_{external} \tag{3.16}$$

$$A_c = A_{cap} = \frac{\pi d_b^2}{4} \tag{3.17}$$

$$A_{rod} = A_c - A_{pistion} = \frac{\pi d_b^2}{4} - \frac{\pi d_{piston}^2}{4} \tag{3.18}$$

In this analysis, P_{cap} and P_{rod} represent the instantaneous pressures in the cap-side and rod-side chambers of the cylinder, respectively. Similarly, A_{cap} and A_{rod} denote the areas of the cap-side and rod-side chambers. The bore diameter of the cylinder is d_b, while d_{piston} is the diameter of the piston.

The force dynamics in the cylinder are governed by the pressure contributions from each chamber, which vary based on which side of the cylinder is pressurized:

- Case 1 ($P_{\text{cap}}A_{\text{cap}}$ dominates): When the cap-side is pressurized, the force contribution from $P_{\text{cap}}A_{\text{cap}}$ drives the motion, opposing the contribution from $P_{\text{rod}}A_{\text{rod}}$.
- Case 2 ($P_{\text{rod}}A_{\text{rod}}$ dominates): When the rod-side is pressurized, the force contribution from $P_{\text{rod}}A_{\text{rod}}$ drives the motion, opposing the contribution from $P_{\text{cap}}A_{\text{cap}}$.

To account for these two scenarios, a switching term, δ, is defined as a piecewise function to represent the direction of the force contributions:

$$\delta = \begin{cases} +1, & \text{if the rod-side is pressurized,} \\ -1, & \text{if the cap-side is pressurized.} \end{cases}$$

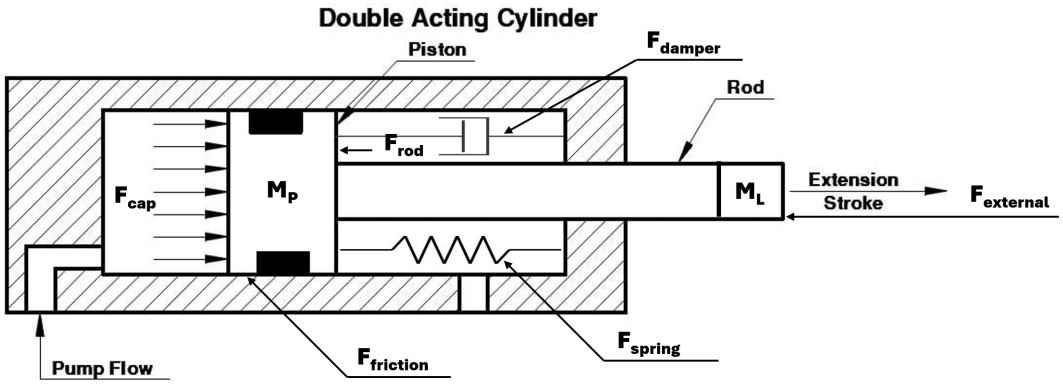

Figure 3.1: Free-body diagram of a double-acting pneumatic cylinder showing chamber forces, rod load, and parasitic effects (spring, damping, friction) [11].

3.1.5 Piston–Joint Coupling and Equivalent Second–Order Model

Combining the chamber pressure dynamics with the free–body diagram in Figure 3.1, the net axial force on the piston–rod assembly is

$$F_{\text{net}}(t) \;=\; \delta(P_{\text{cap}}A_{\text{cap}} - P_{\text{rod}}A_{\text{rod}}) \;-\; F_{\text{friction}}(\dot{x}) \;-\; F_{\text{external}}(\theta), \tag{3.19}$$

where $\delta \in \{+1, -1\}$ indicates which side is commanded (Section 3.1.4), F_{friction} aggregates seal/Coulomb/viscous losses, and F_{external} represents gravity and any distal disturbance reflected to the piston.

Air compressibility around the operating point behaves like a compliant, lightly damped element. Linearizing the chamber energy balance (isothermal or adiabatic) about a nominal pressure P and volume V gives an *air spring* with stiffness

$$k_{\text{air}} \;=\; \frac{\kappa\, A_c^2}{V} \qquad \text{with} \qquad \kappa \;=\; \begin{cases} P, & \text{isothermal,} \\ \gamma P, & \text{adiabatic,} \end{cases} \tag{3.20}$$

and an effective linear air–damping c_{air} (from throttling and leakage) identified experimentally or via small–signal valve models. Using these, the piston coordinate admits a convenient second–order form

$$m_{\text{eq}}\,\ddot{x} \;+\; c_{\text{air}}\,\dot{x} \;+\; k_{\text{air}}\,x \;=\; F_{\text{net}}(t), \tag{3.21}$$

which serves both for control design (PID, hybrid bang–bang/PWM) and for Simulink validation.

Reflection through the chain/sprocket. Let r_{sprocket} denote the effective wrap radius. For the operating range considered (constant effective radius), the piston displacement maps to joint angle as $x \approx r_{\text{sprocket}}\,\theta$ (equivalently, $\theta \approx x/r_{\text{sprocket}}$).

The torque delivered to the joint is

$$\tau_{\text{piston}} \;=\; r_{\text{sprocket}}\, F_{\text{piston}}, \tag{3.22}$$

and the joint dynamics (about the rotation axis) are

$$J_\ell\, \ddot{\theta} \;=\; \tau_{\text{piston}} \;-\; \tau_\ell(\theta, \dot{\theta}), \tag{3.23}$$

where J_ℓ is the lumped rotary inertia about the joint and τ_ℓ includes gravity, friction, and external task torques.

To make (3.21) explicit in x, collect all translating masses rigidly carried by the piston into m_{total} and reflect rotary inertia through the sprocket:

$$m_{\text{eq}} \;=\; m_{\text{piston}} \;+\; m_{\text{total}} \;+\; \frac{J_{\text{total}}}{r_{\text{sprocket}}^2}. \tag{3.24}$$

Here,

$$m_{\text{total}} \;=\; m_{\text{load}} \;+\; \sum_{n=1}^{3} m_n, \qquad J_{\text{total}} \;=\; \sum_{n=1}^{3} \big(J_n + m_n R_n^2\big) \;+\; m_{\text{load}} R_{\text{load}}^2, \tag{3.25}$$

with m_n, J_n the mass and rotary inertia of the nth distal link or payload component, and R_n the distance from the joint axis to its center of mass:

$$R_n(\theta) \;=\; \sum_{i=1}^{n-1} L_i \cos\theta_i \;+\; r_n \cos\theta_n. \tag{3.26}$$

A convenient gravity shorthand uses the mass–weighted radius

$$r_{\text{total}}(\theta) \;=\; \frac{\sum_{n=1}^{3} m_n R_n(\theta) \;+\; m_{\text{load}} R_{\text{load}}(\theta)}{m_{\text{total}}}. \tag{3.27}$$

External force reflected to the piston. Gravity and distal disturbances appear at the joint as a torque

$$\tau_g(\theta) \;=\; m_{\text{total}}\, g\, r_{\text{total}}(\theta) \;+\; \tau_{\text{dist}}(\theta), \tag{3.28}$$

which reflects to an equivalent piston force

$$F_{\text{external}}(\theta) \;=\; \frac{\tau_g(\theta)}{r_{\text{sprocket}}}. \tag{3.29}$$

Substituting (3.29) into (3.19) and then into (3.21) yields a compact, control–ready model in x that captures (i) pressure–generated drive force, (ii) air–spring/air–damping effects, (iii) reflected inertia, and (iv) gravity/disturbance loading.

Summary of constitutive forces (for simulation). For clarity in implementation,

$$F_{\text{spring}} = k_{\text{air}}\, x, \qquad F_{\text{damper}} = c_{\text{air}}\, \dot{x}, \tag{3.30}$$

$$F_{\text{drive}} = \delta(P_{\text{cap}}A_{\text{cap}} - P_{\text{rod}}A_{\text{rod}})\,, \qquad F_{\text{ext}}(\theta) = \frac{m_{\text{total}}\, g\, r_{\text{total}}(\theta) + \tau_{\text{dist}}(\theta)}{r_{\text{sprocket}}}. \tag{3.31}$$

The closed–loop plant used in Simulink thus follows (3.21) with m_{eq} from (3.24) and k_{air} from (3.20). This second–order representation (with input F_{drive} and output x) is directly compatible with the PID and hybrid bang–bang/PWM controllers developed in the next sections.

3.2 Robotic Arm Kinematics

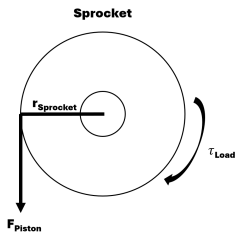

Figure 3.2: Sprocket FBD showing chain tension from the piston acting at the pitch radius and the corresponding torque balance with the external load.

This section formulates the manipulator kinematics using the standard Denavit–Hartenberg (DH) convention. The mechanism is modeled as a planar, 3-DoF serial

chain with two base rotary stages (the stacked pneumatic turntables) followed by the chain/sprocket-driven rotary joint. The piston displacement x is mapped to the third revolute angle via the constant effective sprocket radius r_{sprocket} (Section 5.1):

$$\theta_3 \;\approx\; \frac{x}{r_{\text{sprocket}}} \quad \text{(equivalently, } x \approx r_{\text{sprocket}}\,\theta_3\text{)}.$$

Let the link lengths be $L_1, L_2, L_3 > 0$ measured along the planar $x-y$ workspace, with joint variables $\boldsymbol{q} = [\theta_1, \theta_2, \theta_3]^\top$ defined positive by right-hand rule about $\hat{z}$.

Standard DH table (planar $R-R-R$ chain). Using the standard DH convention, assign frames so that all joint axes are $\hat{z}$-parallel and the links lie in the $\hat{x}$-directions. The parameters are:

Link i	a_i	α_i	d_i	θ_i
1	L_1	0	0	θ_1
2	L_2	0	0	θ_2
3	L_3	0	0	θ_3

With $\alpha_i = 0$ and $d_i = 0$, each homogeneous transform is

$$^{i-1}T_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i & 0 & a_i\cos\theta_i \\ \sin\theta_i & \cos\theta_i & 0 & a_i\sin\theta_i \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad i = 1, 2, 3.$$

The forward pose is $^0T_3 = {}^0T_1\,{}^1T_2\,{}^2T_3$.

Forward kinematics (end-effector position in the plane). Let $c_k \cos\theta_k$, $s_k \sin\theta_k$, and define cumulative angles $\phi_1 = \theta_1$, $\phi_2 = \theta_1+\theta_2$, $\phi_3 = \theta_1+\theta_2+\theta_3$. The end-effector position (x_e, y_e) in the base frame is

$$x_e(\boldsymbol{q}) = L_1\cos\phi_1 + L_2\cos\phi_2 + L_3\cos\phi_3, \tag{3.32}$$

$$y_e(\boldsymbol{q}) = L_1\sin\phi_1 + L_2\sin\phi_2 + L_3\sin\phi_3, \tag{3.33}$$

and the end-effector orientation about $\hat{z}$ is $\psi(\boldsymbol{q}) = \phi_3$. When the third joint is driven by the piston,

$$\theta_3 \;=\; \frac{x}{r_{\text{sprocket}}}, \qquad \Rightarrow \quad \phi_3 \;=\; \theta_1 + \theta_2 + \frac{x}{r_{\text{sprocket}}}.$$

Analytic Jacobian (planar). The velocity mapping $\dot{\boldsymbol{p}} = J(\boldsymbol{q})\,\dot{\boldsymbol{q}}$, with $\boldsymbol{p} = [x_e, y_e, \psi]^\top$, uses the planar Jacobian:

$$J(\boldsymbol{q}) = \begin{bmatrix} -L_1 \sin\phi_1 - L_2 \sin\phi_2 - L_3 \sin\phi_3 & -L_2 \sin\phi_2 - L_3 \sin\phi_3 & -L_3 \sin\phi_3 \\ L_1 \cos\phi_1 + L_2 \cos\phi_2 + L_3 \cos\phi_3 & L_2 \cos\phi_2 + L_3 \cos\phi_3 & L_3 \cos\phi_3 \\ 1 & 1 & 1 \end{bmatrix}.$$

If one prefers to use piston displacement directly as the third generalized coordinate, define $\tilde{\boldsymbol{q}} = [\theta_1, \theta_2, x]^\top$. Since $\theta_3 = x/r_{\text{sprocket}}$,

$$\dot{\theta}_3 = \frac{1}{r_{\text{sprocket}}}\dot{x}, \qquad \Rightarrow \quad \tilde{J}(\theta_1, \theta_2, x) = \begin{bmatrix} J_{(:,1)} & J_{(:,2)} & \dfrac{1}{r_{\text{sprocket}}} J_{(:,3)} \end{bmatrix},$$

where $J_{(:,k)}$ denotes the k-th column of J. This $\tilde{J}$ maps $[\dot{\theta}_1, \dot{\theta}_2, \dot{x}]^\top$ to $[\dot{x}_e, \dot{y}_e, \dot{\psi}]^\top$.

Inverse kinematics (IK) remarks. For the planar 3R case, a common strategy is: (i) choose a desired wrist point (x_w, y_w) for the first two links by subtracting the third link's contribution along ψ, (ii) solve the 2R IK for (θ_1, θ_2) using the law of cosines, and (iii) set $\theta_3 = \psi - \theta_1 - \theta_2$ (or $x = r_{\text{sprocket}}\theta_3$ for piston space). Multiple solutions (elbow-up/down) and singular loci occur where $\det(J) = 0$, e.g., when links align collinearly.

Notes on modeling fidelity. The DH model above assumes (i) the chain/sprocket maintains a constant effective radius over the operating range, (ii) the stacked turntables behave as an equivalent single yaw (their coordination is handled in control), and (iii) the transmission is rigid and lossless (Section 5.1). When large

wrap-angle variations, compliance, or backlash are non-negligible, the mapping $\theta_3 \leftrightarrow x/r_{\text{sprocket}}$ becomes configuration- and load-dependent; in that case, either a calibrated $r_{\text{eff}}(\theta_3)$ should replace r_{sprocket} or a small joint-offset state should be estimated and compensated in control.

CHAPTER 4

DESIGN

This chapter presents the design process of the Off Riser Pneumatic Manipulator (ORPM), a robotic manipulator developed to safely and efficiently retrieve samples from the interior of high-risk waste storage tanks at the Hanford Site. The system's design is driven by strict environmental, spatial, and safety constraints, requiring a thorough evaluation of materials, structural components, and mechanisms for remote operation. The chapter outlines the conceptual and detailed design phases, highlighting considerations for material selection based on radiation resistance, corrosion tolerance, and mechanical strength. Finite Element Analysis (FEA) was used to validate the structural integrity and performance of critical components under expected loading conditions. Design for Manufacturing (DFM) principles were applied to ensure ease of assembly, reliability in remote environments, and cost-effectiveness. In addition, a kinematic model of the full arm assembly was developed to support control system design and trajectory planning, ensuring precise and repeatable operation within confined tank risers. The results of this comprehensive design effort form the basis for the prototype implementation and testing discussed in subsequent chapters.

4.1 FEA Analysis

As a critical component of the mechanical design process, finite element analysis (FEA) was employed to validate and refine the structural integrity of the Off-Riser Sampler System. This analysis was built upon the foundational weight optimization work conducted previously[6], whose contributions guided key material and geometric selections. Furthermore, experimental feedback gathered during the sum-

mer deployment phase of the system offered valuable real-world performance data, highlighting areas of operational stress and system behavior under typical field conditions. The purpose of this FEA was to assess the theoretical performance limits of the final design, with a focus on stress distribution, deformation, and overall structural resilience. Using this simulation-based approach, critical loading scenarios were predicted and confirmed whether the system could safely operate within anticipated constraints. The FEA results were then cross-referenced with experimental outcomes, offering an additional layer of validation and confidence in the system's mechanical performance. This subsection presents the analytical methodology, key assumptions and boundary conditions, and a discussion of the resulting stress and deformation profiles. The findings not only demonstrate the robustness of the final design but also support its readiness for real-world deployment and extended operational use.

4.1.1 Chain Adapter

Chain Adapter Under Tension

The chain adapter was evaluated under chain pre-tension and operational pull using a linear-elastic FEA model for carbon steel. The piston-side region was constrained as a rigid attachment to the actuator rod, and loads were applied as tensile pull through the chain link holes to reflect the actual load path. Geometry and constraints follow the design shown in Figures 4.1–4.2. For a properly tensioned ANSI roller chain, we used a representative pre-tension of $T_0 = 200\,\mathrm{N}$ (typical practice is $T_0 = \alpha W_{\mathrm{rated}}$, $\alpha \in [0.05, 0.10]$) and superposed a worst-case operating increment to form $T_{\max}$.

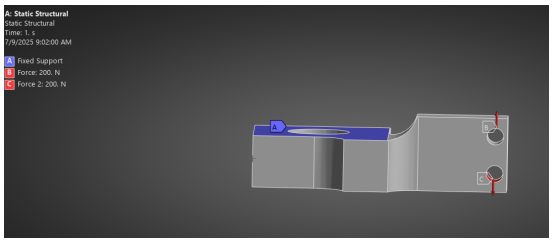

Figure 4.1: Chain adapter boundary conditions for tensioned chain

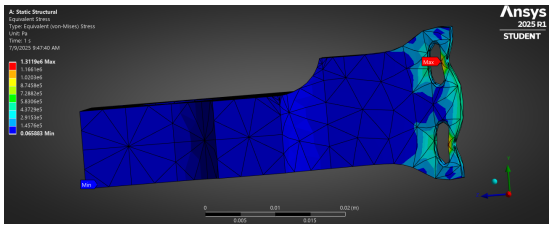

Figure 4.2: Stress results for tensioned chain

The analysis produced a peak von Mises stress of $\sigma_{\mathrm{vm,max}} = 1.3119 \times 10^6\,\mathrm{Pa}$ ($\approx 1.312\,\mathrm{MPa}$) at the fillet adjacent to the chain-hole load path, which is far below the yield strength of common carbon steels ($\sigma_y \gtrsim 250\,\mathrm{MPa}$). Deflections at the chain interface were small, supporting the rigid, linear transmission assumption used for control (Section 5.1).

In summary, with the piston interface modeled as rigid, loads applied through the chain holes, and the chain properly tensioned via the in-line tensioner, the adapter exhibits ample strength margin and stiffness. These results justify treating piston displacement as mapping to joint angle with negligible lost motion within the operating range.

Chain Adapter Under Tension and end effector load

The stress evaluation with a tensioned chain and an equivalent bending moment (representing manipulator and end-effector weight) is summarized in Figure 4.3. This configuration captures the realistic tendency of the payload to induce additional bending at the chain interface during sampling, elevating stress near fillets and around the chain-hole bearing surfaces while preserving the actual load path.

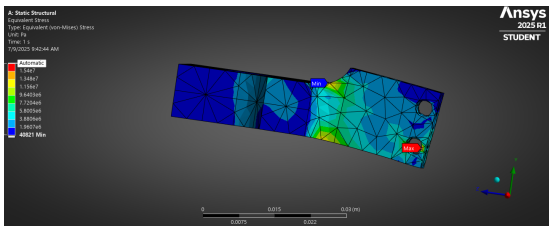

Figure 4.3: Stress under tension + payload moment.

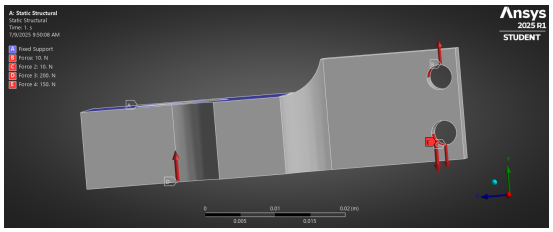

Figure 4.4: Boundary conditions: tension + moment.

Figure 4.3 was constructed with chain pre-tension applied as tensile pull through the link holes, a gravity-equivalent tip moment superposed about the adapter's neutral axis, and a rigid constraint at the piston-side attachment to represent the stiff actuator connection. The applied forces also include the preload of the clamping bolt and the estimated downward load from the manipulator/end effector so that combined effects on the adapter are reflected in the boundary conditions.

Under the combined chain tension, bolt preload, and payload-induced moment, the peak von Mises stress was approximately $1.54 \times 10^7\,\mathrm{Pa}$ ($\approx 15.4\,\mathrm{MPa}$), which remains well below the yield strength of the carbon steel. Corresponding interface deflections are small, so the adapter provides sufficient stiffness to uphold the rigid, linear transmission assumption for control, while proper in-line chain tensioning maintains a repeatable $x \to \theta$ mapping across operating conditions.

4.1.2 Pneumatic Cylinder

This section validates the preliminary design choice to use the pneumatic cylinder itself as a primary load-carrying member of the manipulator. The cylinder is stainless steel and, by design, carries axial compression during operation. Two effects govern its stability: (i) the classical Euler column capacity of a thin-walled tube and (ii) a pressure-stabilization effect due to internal operating pressure, which in-

troduces beneficial membrane stresses in the shell. Intuitively analogous to a sealed beverage can, the pressurized cylinder exhibits substantially greater resistance to axial buckling than the same shell when vented; once depressurized or "opened," the added stabilization vanishes and the member reverts to its unpressurized column capacity.

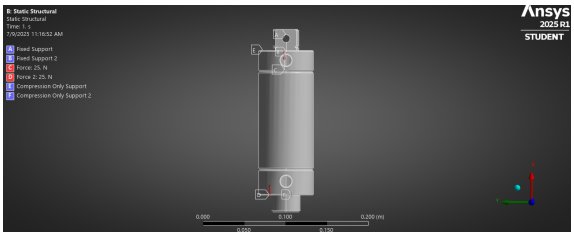

Figure 4.5: Boundary conditions of Pneumatic Cylinder

Under pressure the cylinder exhibited a substantially elevated buckling margin consistent with pressure-stabilized shell theory; nonlinear runs confirmed increased collapse load and delayed mode localization. Peak von Mises stresses remained below the stainless-steel yield limit, and axial shortening stayed within alignment tolerances, supporting use of the cylinder as a structural member when pressurized. Because only the pressurized case was evaluated, the design treats internal pressure as a required pre-stress: pressure monitoring and interlocks are recommended so that compressive loads demanded during operation remain within the validated pressurized capacity and the rigid, linear transmission assumptions used for control remain valid.

4.2 Design Improvements and Updates

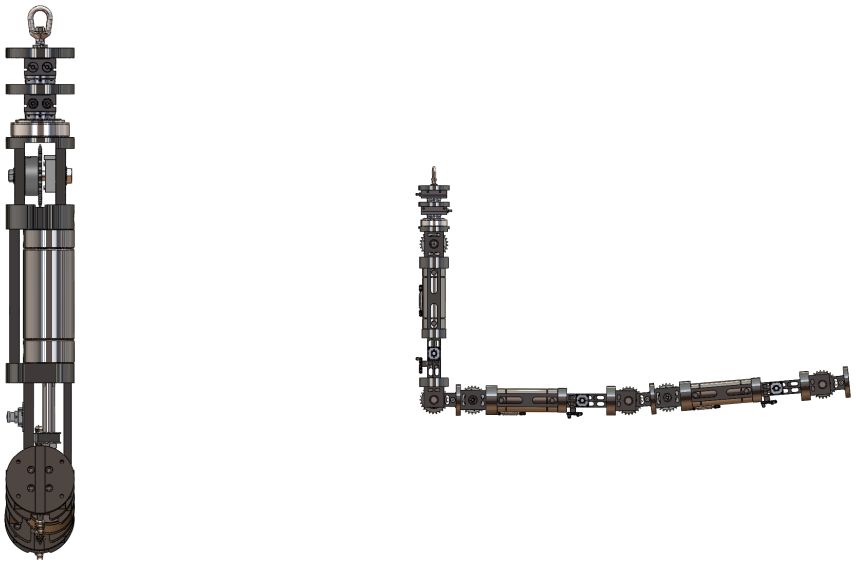

Figure 4.6: Fully assembled manipulator in the same configuration shown from orthogonal views (front and side).

As shown in Figure 4.6, the arm remains fully modular: the linear (piston) stage, stacked rotary turntables, chain adapter, and sensor shrouds mount as discrete subassemblies using common bolt patterns and dowel datums. Harnessing, pneumatic manifolds, and shields are routed on removable trays, so any module can be swapped without disturbing adjacent stages. This preserves bench-top serviceability while accommodating the added rotary joint and sensor package, and maintains clear cutter/tool access for CNC-produced parts and straightforward rework.

To support safe handling and future field deployment, the assembly incorporates purpose-built hoist points at the base frame and mid-span bracket, each gusseted and through-bolted into primary load paths for two-leg bridle lifts about the assembly center of gravity. These interfaces double as mounting locations for forthcoming stabilization hardware (e.g., yokes, guide rails, snubbers) and for standardized de-

ployment frames; secondary bosses along the base plate align with cart/test-stand patterns, and cable/pneumatic strain-relief bars adjacent to the hoist points prevent load transfer into connectors during lifts.

4.2.1 Rotational Joint Addition

To add a rotational DoF, we employ compact aluminum rotary pneumatic turntables as a serial stage at the base of the prismatic axis as shown in Figure 4.7. Each unit is a vane-type actuator limited to 180°, so two identical units are stacked out of phase (zeros offset by 180°). Sequencing the pair yields full orientation coverage: one table recenters while the other advances, avoiding hard-stop impacts and distributing torque demand.

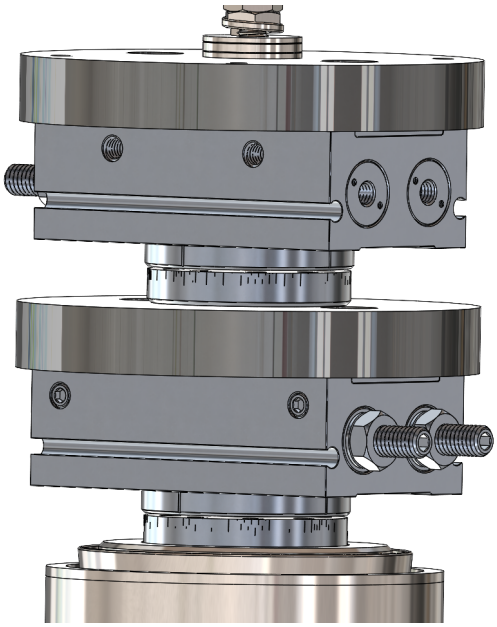

Figure 4.7: Stacked precision rotary tables used for compound orientation control at the base of the manipulator.

Because the turntables cannot safely accelerate the fully extended manipulator at far-from-neutral orientations (peak reflected inertia and gravity moment), large

reorientations follow a return-to-neutral rule: retract the prismatic axis toward a neutral extension, re-center the more offset table, execute the required $\Delta\Theta$ primarily with the near-neutral stage, then re-extend and settle with the same deadband/PID shaping used for the piston. Under this policy the rotational joint respects torque and stop limits, preserves the rigid-link assumption of Section 5.1, and achieves piston-parity in control behavior.

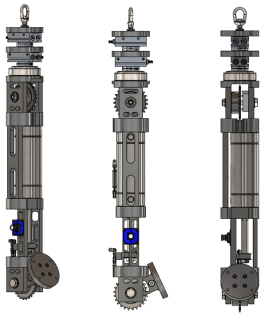

Figure 4.8: Full rotational joint member at different positions

4.2.2 Sensor Integration

Three sensing modalities are integrated to observe state and validate control: a 1D LiDAR embedded within the cylinder mount to measure piston displacement, a rotary encoder on the sprocket to measure joint angle, and a camera for visualization and position checks. The LiDAR is recessed behind approximately one to two inches of surrounding metal (with a clear line-of-sight to a reflective target on the moving member), providing substantial shielding. The encoder is mounted coaxially with the drive for high-resolution angle measurement with minimal added inertia. The camera views the work envelope for alignment and qualitative verification. Under the nominal rigid, linear mapping between piston displacement and joint rotation, the controller checks for consistency between LiDAR-based displacement and encoder-based angle and blends them with a lightweight estimator when

they agree; if they diverge, the logic flags possible transmission compliance/backlash or a sensor fault and degrades gracefully (wider deadbands, reduced approach speeds, guarded moves).

Environmental constraints drive a fail-operational design. The camera is treated as non-mission-critical and expected to fail under radiation; it is used primarily for setup and qualitative post-checks. The rotary encoder is selected for radiation tolerance and is expected to behave similarly to a potentiometer in the short term (minor drift), while the LiDAR's metal shroud provides sufficient attenuation for the operational window. If any onboard sensor is rejected (timeouts, out-of-range, excessive noise) the system remains operable with no onboard feedback: the controller switches to open-loop, rate-limited sequences with conservative deadbands, and operators use external tank visualization to guide sampling (arcade "claw-machine" style). Motion planning in degraded modes returns through neutral/low-inertia poses, and interlocks prevent exceeding validated travel and force envelopes.

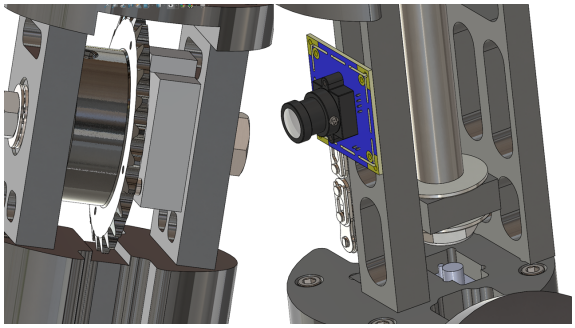

Figure 4.9: Integrated sensor package mounted on the joint module: camera board and auxiliary electronics.

4.2.3 Design for Manufacturing

All custom components are designed for direct CNC manufacture from standard plate, bar, or round stock using conventional 3-axis milling and turning. Geometry

is prismatic (pockets, bores/slots, counterbores, simple seal grooves) with clear tool access, few setups, and no 5-axis-only features. Fillet radii, minimum wall thicknesses, and hole patterns are sized for common cutters and off-the-shelf tooling. Figure 4.10 highlights stock sizes and cutter approach directions, demonstrating that every part is directly machinable.

Stainless parts are specified to ASTM A240[22]; carbon-steel brackets and adapters to ASTM A36 [23](or ASTM [24] where higher strength is required); and critical fasteners/pressure-boundary hardware to ASTM A193/A194[25, 26]. Aluminum custom parts are specified to ASTM B209 and are finished with black anodize (Type II, Class 2) for corrosion/wear resistance and low reflectance; steel components receive a black oxide finish where appropriate. An exterior clear acrylic shell (impact-modified PMMA[27]) encloses the assembly to provide added environmental protection (splash, particulate, incidental debris) and visual inspection capability; standoffs and seals maintain service access without imparting undue loads to mounted hardware. Where applicable at the component level, purchased pneumatic hardware conforms to ISO 15552[28] (cylinder dimensions/interfaces), and local manipulator pneumatics observe essential clauses of ISO 4414[29] limited to the manipulator (e.g., local relief and isolation).

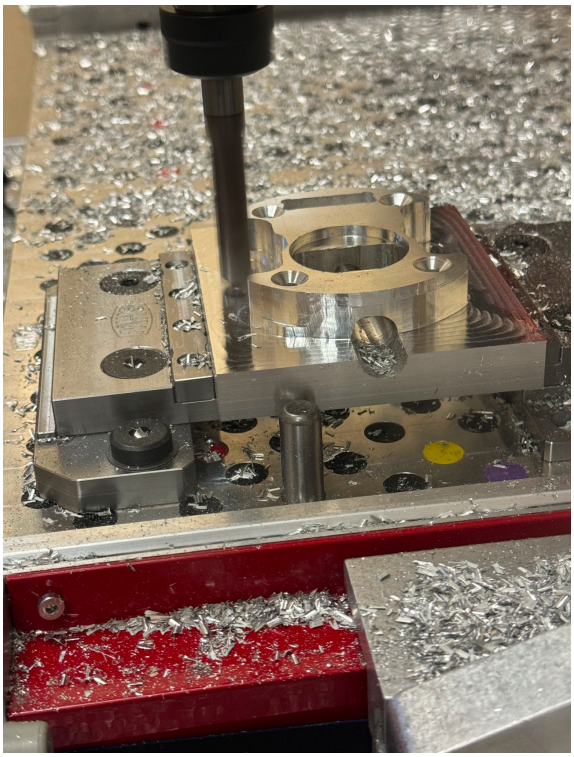

Figure 4.10: Manufacturing of an AL-6061 component on a 3-axis CNC milling setup.

Design for Assembly (DFA) Considerations

To support assembly in hot/contaminated cells and simplify field service, we adopted the following DFA practices:

- **Fastener minimization and standardization:** all external fasteners limited to two hex sizes; captive nuts and washer-head screws used where glove dexterity is limited.
- **Decontamination-friendly geometry:** avoided blind holes and deep crevices in external surfaces; radiused internal corners and through-holes preferred to enable spray/rinse decon.

- **Material/process selection:** external components in 6061-T6 or 304/316 stainless with brushed finishes; avoid paints where flaking may occur under radiation and abrasion.
- **Fastener control:** prevailing-torque locknuts and threadlocker only in sealed volumes; safety-wire on critical joints accessible to tools with limited reach.

CHAPTER 5

CONTROLS

Placing the equation of motion into state-space form, we simplify the system into a first-order differential equation. The state variable, X, is represented by:

$$X = \begin{bmatrix} x \\ \dot{x} \end{bmatrix} \quad \therefore \quad \dot{X} = \begin{bmatrix} \dot{x} \\ \ddot{x} \end{bmatrix} \tag{5.1}$$

$$\ddot{x} = \frac{1}{m}(F_{net} - c_{air}\dot{x} - k_{air}x) \tag{5.2}$$

State-space is representation of the system is: $\dot{X} = AX + Bu$. Where A is the system matrix, B is the input matrix, and $u = F_{net}$ is the input.

$$\dot{X} = \begin{bmatrix} \dot{x} \\ \ddot{x} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -k_{air}/m & -c_{air}/m \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} F_{net} \tag{5.3}$$

And the output can be represented by: $y = CX + Du$. Where C and D specify the output variable.

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} \tag{5.4}$$

5.1 Output Variable and Modeling Assumptions

The desired controlled output in this work is the linear position of the pneumatic cylinder piston, denoted $x(t)$. For the scope of this paper, the transmission between the piston and the joint—comprising the chain drive and mechanical adapter—is assumed rigid and its behavior linear and predictable. Under these assumptions, a displacement of the piston produces an instantaneous, lossless angular displacement at the driven joint:

$$\theta(t) \; = \; \theta_0 \; + \; k\, x(t), \tag{5.5}$$

where $\theta(t)$ is the joint angle, θ_0 is a geometric offset, and $k > 0$ is a constant kinematic gain set by the sprocket/pulley radii and adapter geometry. Thus, regulating $x(t)$ is equivalent to regulating $\theta(t)$, and the controller can be designed and analyzed entirely in the piston-coordinate domain.

5.1.1 Assumptions

- Rigid transmission: Chain, adapter, and mounting interfaces exhibit negligible compliance and backlash; no elastic stretch or slack is modeled.
- Proper chain tension: The above rigidity and linearity assumptions hold only if the chain is correctly tensioned using the in-line tensioner described in the Introduction[8]; insufficient tension would introduce backlash, lost motion, and a state-dependent gain [30].
- Linear kinematics: The $x \to \theta$ mapping is well-approximated by a constant gain k over the operating range (no geometric nonlinearity or wrap effects within this range).
- Lossless coupling: Energy losses due to friction, hysteresis, or micro-slip in the chain/adapter are neglected; no lost motion is assumed.
- Repeatability: The mapping $x \mapsto \theta$ is repeatable across cycles (no thermal growth or wear effects over the time scales considered).

5.1.2 Implications for control.

Under these conditions, the controller regulates $x(t)$ directly (Sections 5.4–5.5); setpoints, errors, and performance metrics are all expressed in piston coordinates. Because $\theta = \theta_0 + kx$, any tracking specification in angle space (e.g., $\theta^\star$) maps to an

equivalent piston setpoint $x^\star = (\theta^\star - \theta_0)/k$. This simplification allows the pneumatic actuation and fluid dynamics to be addressed without additional nonlinearities from the transmission.

These assumptions are valid for small-to-moderate excursions where the chain remains engaged on a constant effective radius, and where proper chain tension is maintained by the in-line tensioner. If significant compliance, backlash, tension drift, or nonlinear geometry appears (e.g., varying wrap angle, load-dependent stretch), the effective gain k becomes state- and load-dependent, and the controller may require gain scheduling, feedforward compensation, or an inner-loop estimator to maintain performance.

5.2 URDF-Driven Visualization and Dynamics Analysis in RViz and Isaac Sim

To verify kinematics and self-collision behavior prior to hardware testing, I exported the manipulator model to the Unified Robot Description Format (URDF) with consistent link frames, joint limits, and both visual and collision geometries. The URDF was launched in RViz via robot_state_publisher; joint_state_publisher_gui was used to sweep each degree of freedom and observe real-time joint motion. RViz's RobotModel and TF displays provided immediate feedback on frame alignment, parent–child connectivity, and axis conventions, which I used to validate that the kinematic chain matched the intended design. Enabling the collision overlay allowed rapid inspection of contact hulls and identification of potential self-collisions near joint limits. This RViz pass established that visual meshes, collision meshes, and inertial tags were co-registered, and that limit values prevented singular or mechanically infeasible postures.

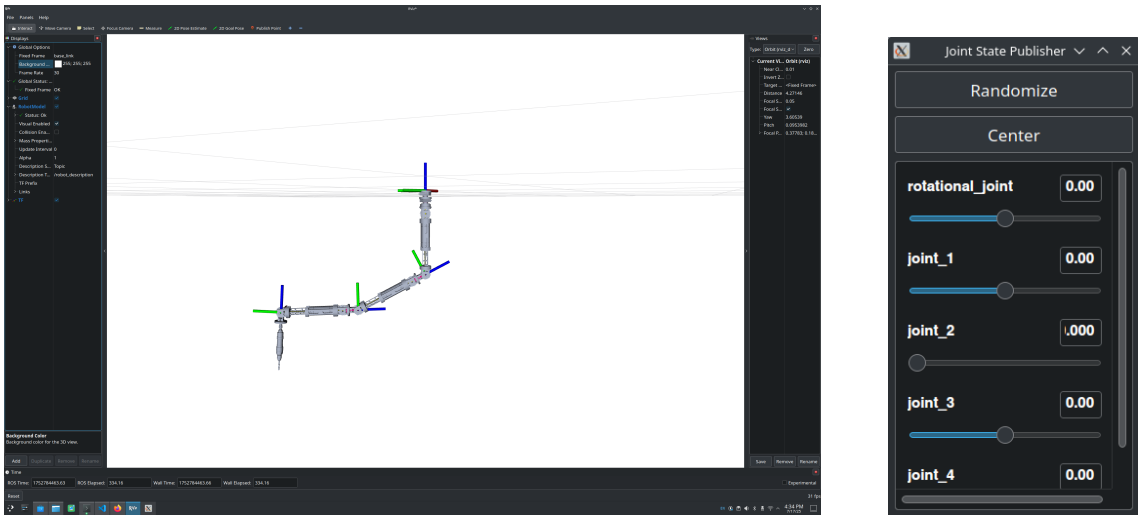

Figure 5.1: RViz scene used to verify the ORSS URDF and kinematics, including drill end-effector and joint-space actuation controls.

For dynamic behavior under load, the same URDF was imported into NVIDIA Isaac Sim via the URDF importer to generate a USD articulation with PhysX dynamics. Following verification of link masses and principal moments from the URDF inertial tags, realistic material properties (density, friction, restitution) were assigned, gravity was enabled, and physics solver parameters (iterations and substeps) were configured to ensure stable contact handling and articulation integration. Representative motion profiles were then executed using the articulation controller to quantify end-effector deviation arising from self-weight and motion-induced loads. Logged signals included joint positions and velocities, commanded efforts, and measured torques; these were compared with the commanded trajectories to compute tracking-error envelopes.

The simulations informed both structural and control design decisions. Static deflection under self-weight was used to estimate required actuator strength and stiffness margins, whereas dynamic responses (overshoot and oscillation) guided controller tuning. Gravity compensation/feedforward was incrementally incorpo-

rated, joint-space proportional–derivative gains (drive stiffness and damping) were adjusted, and conservative velocity and acceleration limits were applied. The objective was to minimize elastic deflection, overshoot, and limit-cycle oscillations while preserving tracking accuracy. Contact and penetration reports in Isaac Sim were cross-checked against self-collision configurations identified in RViz to verify that the dynamic envelope remained within safe bounds.

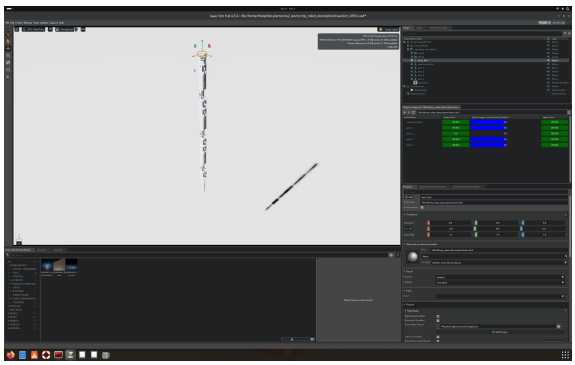

Figure 5.2: NVIDIA Isaac Sim workspace: URDF-imported manipulator with PhysX settings and live plots for kinematic/physics verification.

In summary, RViz provided a fast, interactive environment to validate kinematics, frames, and collision geometry from the URDF, while Isaac Sim supplied physics-accurate insight into how the manipulator behaves under its own weight and during motion. The combined workflow de-risked hardware bring-up by surfacing kinematic, structural, and control issues in simulation, and produced controller setpoints (stiffness, damping, and feedforward terms) sized to the observed torque demands and error envelopes for the intended duty cycle.

5.3 MATLAB/Simulink Simulation of the Pneumatic Actuator

This section documents the MATLAB/Simulink model used to study the closed–loop behavior of the pneumatically actuated ORSS joint. The model is implemented in Simulink with Simscape Fluids and represents the full pneumatic circuit, load dynamics, and sensing/actuation interfaces. The goals of the simulation were to: (i) verify that the directional valve and reservoir sizing provide adequate flow for the commanded trajectories, (ii) quantify load displacement and steady-state error under representative forces, and (iii) generate time histories for controller tuning and comparison to bench tests.

5.3.1 Model Architecture

Figure 5.3 shows the top-level Simscape network. A 5/3 directional control valve routes supply and return flow to a double-acting cylinder through line volumes and restrictions that emulate hoses and fittings. Check valves, accumulators/reservoirs, and line losses are included to capture transient compliance and pressure lag. The electrical/logic portion converts a normalized control input into spool position. Cylinder motion drives the prismatic joint that represents the ORSS axis under study, while the Position sensor block publishes actuator displacement for logging and comparison matching the system output shown Equation 5.4.

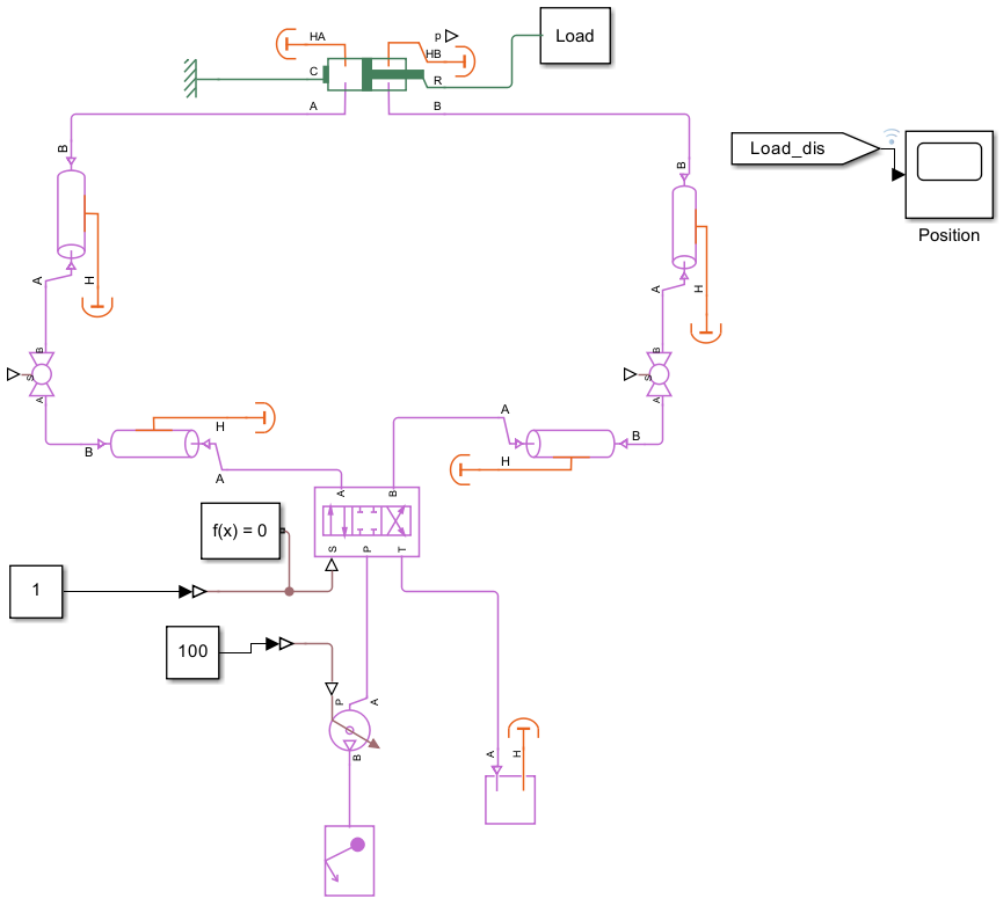

Figure 5.3: Top-level Simscape Fluids model of the ORSS pneumatic axis, including the 5/3 directional valve, line dynamics, accumulators, and the double-acting cylinder with position sensing.

The load path is shown in Figure 5.4. The actuator rod connects to a translational mass–spring–damper element that represents the reflected inertia, structural compliance, and passive damping of the mechanism at the working point. An external Load Force input allows application of constant or time-varying disturbances (e.g., gravity components or contact), while the Load_dis probe outputs displacement for downstream analyses.

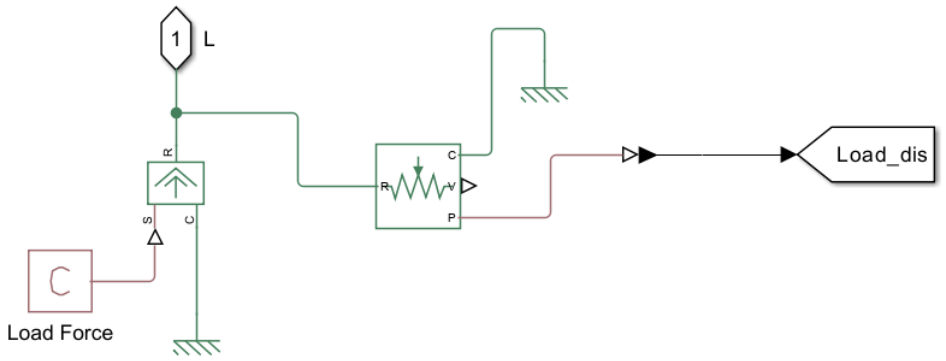

Figure 5.4: Load dynamics and measurement: translational mass–spring–damper connected to the actuator rod with provision for external disturbance force and logged displacement signal `Load_dis`.

5.3.2 Parameterization and Assumptions

Valve metering areas, flow coefficients, and spool travel were set from datasheet nominal values; hose lengths and diameters reflect the test stand routing. Chamber volumes include dead volume plus stroke-dependent volume, and air is modeled as a compressible ideal gas with temperature dynamics neglected (isothermal assumption) for the time scales of interest. Leakage paths are represented by small orifices to ground. The load mass and equivalent stiffness/damping were identified from static deflection and ring-down tests and then refined to match measured settling envelopes.

5.3.3 Simulation Setup

Simulations were run with a variable-step, stiff solver and tight relative tolerance to capture choked/un-choked transitions and contact-like pressure spikes without

numerical chatter. Each run used a step-and-hold position command followed by a multi-segment trajectory to exercise both flow-limited and compliance-limited regimes. Logged signals included chamber pressures, valve spool position, mass flow rates, and Load_dis. To support controller work, the command/response pair was exported for frequency-domain fits (empirical second-order plus delay) and for time-domain error metrics.

5.4 Bang-Bang Control for a Pneumatic Actuator with 5/3 Valves

The manipulator's linear joint is driven by a double-acting pneumatic cylinder with two chambers (cap and rod), each plumbed to an independently controlled 5/3 directional solenoid valve. With digitally commanded spool states—pressurize, exhaust, or closed—the net piston force is selected from a small set of discrete levels, making an on/off (bang–bang) strategy a natural fit. In this design, valve states are chosen directly from the sign of the position error, with a small deadband to halt near the target rather than computing a continuous control signal.

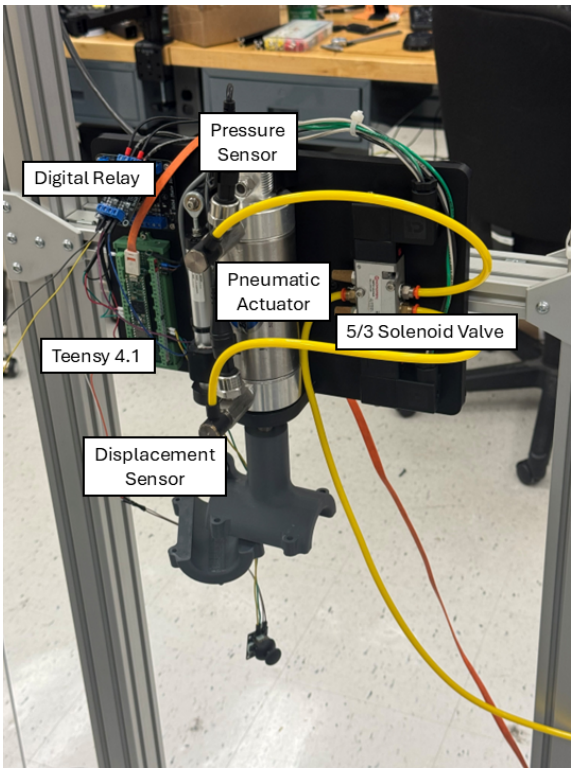

Figure 5.5: Bench setup used to validate bang–bang control: Teensy 4.1 with relay interface, displacement and pressure sensing, 5/3 directional valve, and pneumatic actuator.

Let the reference be a step $r(t) = r_0 H(t)$ and define the position error $e(t) = r(t) - x(t)$. The firmware uses the following decision law:

$$|e| < \text{Buffer} \Rightarrow \text{HALT}, \quad e > 0 \Rightarrow \text{CAP}, \quad e < 0 \Rightarrow \text{ROD}.$$

Here, Buffer is a deadband that suppresses chatter by closing both chambers when the target is approached. The actions map to relay states as:

- CAP (extend): cap pressurize, rod exhaust $\Rightarrow$ SOLENOID_CAPSIDE_RELAY=HIGH, SOLENOID_RODSIDE_RELAY=LOW.
- ROD (retract): rod pressurize, cap exhaust $\Rightarrow$ LOW/HIGH.
- HALT (hold): both closed $\Rightarrow$ LOW/LOW.

Operationally, a step in $r(t)$ produces a saturated push in the direction of the error (CAP for $e > 0$, ROD for $e < 0$). As the piston nears the setpoint, the deadband triggers HALT, trapping chamber pressures so pneumatic/compliant damping dissipates kinetic energy and limits overshoot. Independent chamber control also exploits the area asymmetry (cap > rod) to generate large differential pressures for rapid acceleration and braking.

This controller is intentionally discrete and step-like: the commanded action changes only when the sign of e (outside Buffer) changes. The HALT region acts as a hold/settle mode that reduces the limit cycles typical of pure on/off control while preserving the fast rise time of bang–bang actuation. As carried mass increases, inertia and approach momentum grow for the same available differential pressure, so deceleration into HALT becomes slower. Because HALT traps chamber pressures, the compressible air behaves like a spring; higher momentum stores and releases more energy, leading to deeper penetration past the setpoint, greater overshoot, and oscillation before settling. A practical mitigation is to increase Buffer so the system enters HALT earlier, providing more time/distance to dissipate energy and thereby reducing overshoot and oscillation. The trade-off is reduced positioning precision, since a larger Buffer widens the acceptable final error band. (Additional mitigations that preserve precision include slowing the approach near the target, adding a small velocity term or gate to the switch, or applying gravity/feedforward to offset weight.)

5.4.1 Firmware Excerpt (exact control logic).

```
/* Decision: deadband around zero, else drive by sign(error).
   error = setpoint - position
```

```
*/
if (fabs(error) < Buffer) {
  action = HALT;
} else if (error > 0) {
  action = CAP;   // extend: cap pressurize, rod exhaust
} else {          // error < 0
  action = ROD;   // retract: rod pressurize, cap exhaust
}


void actSolenoid(int state) {
  switch (state) {
    case CAP:
      digitalWrite(SOLENOID_CAPSIDE_RELAY, HIGH);
      digitalWrite(SOLENOID_RODSIDE_RELAY, LOW);
      break;
    case ROD:
      digitalWrite(SOLENOID_CAPSIDE_RELAY, LOW);
      digitalWrite(SOLENOID_RODSIDE_RELAY, HIGH);
      break;
    default: // HALT
      digitalWrite(SOLENOID_CAPSIDE_RELAY, LOW);
      digitalWrite(SOLENOID_RODSIDE_RELAY, LOW);
  }
}
```

5.4.2 Bang–Bang Controller Results and Analysis

Figures 5.6, 5.7, and 5.8 show a tight causal link between the discrete control $u \in \{-1, 0, +1\}$, the chamber pressures, and piston motion. When $u = +1$, the cap chamber is fed and the rod chamber vented, yielding a positive differential

$$\Delta P(t) = P_{\text{cap}}(t) - P_{\text{rod}}(t) > 0,$$

which accelerates the piston toward the setpoint; switching to $u = -1$ reverses ΔP and pulls the piston back. The position trace exhibits a fast rise followed by overshoot, an undershoot, and a small steady oscillation about the target. These oscillations persist despite the existing deadband, indicating that pneumatic compressibility, valve/flow delays, and inertia allow ΔP to continue evolving after each switch, so momentum carries the piston past the setpoint before the reversed command can dissipate it. The pressure plots confirm this mechanism: step-like changes in the solenoid command produce delayed, ramped changes in P_{cap} and P_{rod}; the sign and magnitude of ΔP closely track the acceleration and the observed overshoot/undershoot sequence. Implications are (i) excellent rise time due to large $|\Delta P|$ under saturation, (ii) lightly damped behavior with a limit cycle set by the deadband and the plant's lags, and (iii) sustained switching near the setpoint, which increases valve wear and pneumatic consumption.

Motivation for a PID regulator.

- Reduced overshoot and settling time: the proportional term tapers commanded effort as $|e| \to 0$, while the derivative term adds velocity feedback to arrest the piston before crossing the setpoint—capabilities not available with discrete switching.

- Elimination of residual bias and limit cycle: the integral term removes steady errors due to friction and load imbalance, enabling regulation within a tight band without dithering inside the deadband.
- Lower switching stress and smoother pressures: a PID output (implemented as valve PWM/duty or a servo-valve command) produces graded ΔP rather than step changes, reducing chatter at the setpoint and decreasing mechanical wear and air usage.

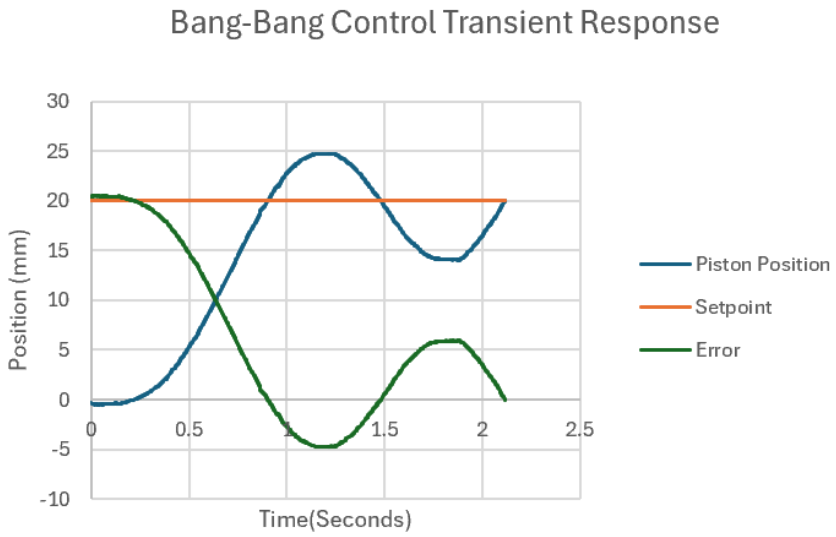

Figure 5.6: Piston position (blue) tracking a constant setpoint (orange) with error (green). The response exhibits fast rise, overshoot, undershoot, and a small steady oscillation about the setpoint, characteristic of bang–bang actuation on a compressible pneumatic plant with a finite deadband.

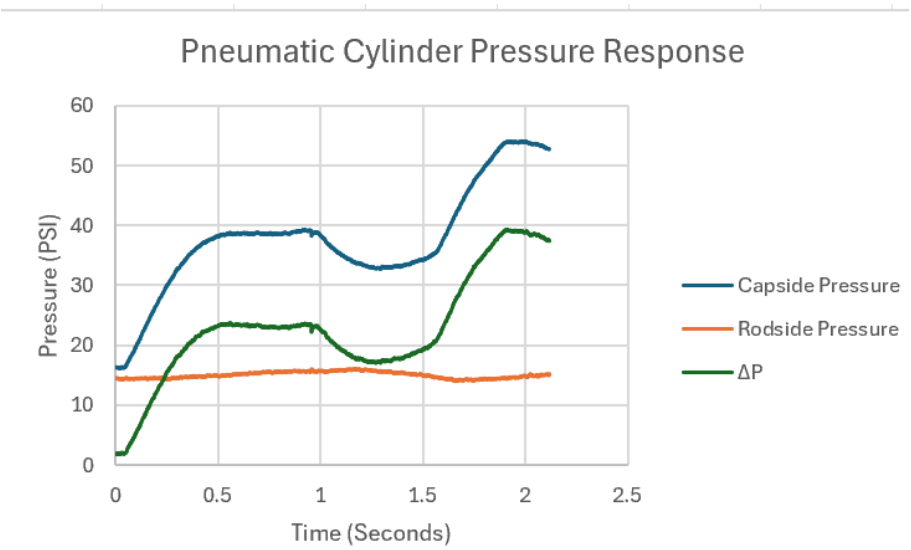

Figure 5.7: Chamber pressures and differential $\Delta P(t) = P_{\text{cap}} - P_{\text{rod}}$. Discrete valve states create step-like changes in chamber supply/venting; the resulting ΔP sign and magnitude explain the acceleration phases that produce the observed overshoot/undershoot.

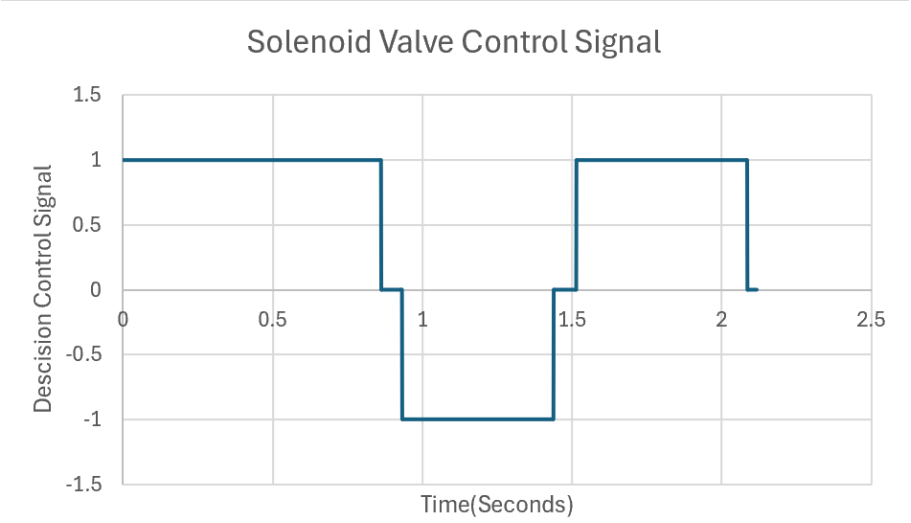

Figure 5.8: Bang–bang control signal $u \in \{-1, 0, +1\}$ showing feed, neutral (deadband), and reverse states. Switches near zero error, together with pneumatic lag and inertia, drive the limit cycle seen in the position response.

5.5 PID Controller

To achieve finer positioning than bang–bang actuation, we evaluated closed-loop PID control using (i) proportional metering valves that modulate flow rate into each chamber and (ii) electronic pressure regulators (EPRs) that command chamber pressure directly. Both approaches are governed by the ideal-gas relation and the same force balance on a double-acting cylinder, but they differ in how the controller influences the chamber states and, consequently, in their precision and robustness.

5.5.1 Proportional Valve PID (flow control).

With proportional valves, the PID loop modulates valve openings to meter inflow to both chambers simultaneously (no deliberate exhausting). As the commanded openings shrink, the chamber producing the larger force dominates and nudges the piston toward the setpoint, yielding high resolution near the target because small flow changes induce small pressure changes. When the error is very small, the effective orifice area is nearly closed, so the mass flow $\dot{m} \to 0$. The trapped air then behaves like a compliant spring–damper: the temporarily "choked" inflow/outflow resists further pressure change and helps dissipate kinetic energy, which can damp overshoot at the setpoint.

The same mechanism imposes a trade-off when the setpoint changes again. Because there is no active exhaust, the total air mass in the two chambers tends to ratchet upward during corrections; with sustained nonzero inflow and near-closed orifices, both p_c and p_r drift toward a maximum $p_{\max}$ set by the supply/regulator. As pressures rise, the opposing chamber presents greater back-pressure, making subsequent motion harder (higher force required to initiate and arrest movement), slowing halts, and increasing the risk of residual bias forces. Practically, this manifests as

sluggish response to new commands and a higher propensity for overshoot/settling oscillations unless integrator anti-windup and conservative gains are applied.

Electronic pressure regulation PID (pressure control). EPRs close the loop at the pressure level. A position–space PID computes a desired differential pressure Δp (or individual chamber targets $p_c^\star, p_r^\star$), and the regulators track those setpoints by adjusting their internal valves, including an exhaust stage. This directly addresses the "pressure ratcheting" observed with proportional flow control: because an EPR can lower chamber pressure on demand, trapped energy does not accumulate during small corrections.

Standard EPRs are downstream devices: they can only reduce pressure from the supply and cannot boost above it, so feasible setpoints lie within $[p_{\mathrm{atm}}, p_{\mathrm{sup}}]$. Operating at lower commanded pressures reduces available actuation force ($F \propto \Delta p$), which weakens disturbance rejection and can slow the final approach under heavier or time–varying loads.

An additional benefit is that one chamber can be held at a constant pressure while the other is varied, shaping Δp by moving just one side. In principle, proportional flow valves could emulate this behavior, but without an active exhaust this typically amplifies the low–pressure constraint: the "constant" side must be held relatively low, further shrinking the achievable Δp. For a manipulator that encounters changing payloads and contact disturbances during sampling, this reduced differential–pressure headroom is significant; maintaining tracking accuracy then requires higher supply pressure, gain scheduling, or a hybrid scheme (e.g., an EPR inner loop with selective bang–bang or metered assist) to recover force when needed.

Two-stage (cascade) PID concept. A cascaded approach can resolve the weaknesses of pure proportional–flow and pure bang–bang schemes. When a new setpoint is issued, chamber pressures can be actively adjusted (vented or increased up to supply) so previously accumulated pressure does not "ratchet" the system, while a proportional valve remains the primary driver of motion. Unlike the step–like action of bang–bang, the proportional valve command can be shaped as a linear, exponential, or logarithmic function of the tracking error to tailor approach speed and damping. Because the piston force ultimately follows the ideal gas law, simultaneously controlling both pressure (via EPRs or pressure feedback) and mass flow (via proportional metering) yields finer precision and better transient control.

A practical compromise is a cascade: an outer position PID computes a desired differential pressure Δp for responsiveness, while an inner pressure PID (realized by EPRs or by fast proportional valves with pressure feedback) tracks p_c and p_r. Near the target, a scheduler reduces $|\Delta p|$ and, if using proportional valves, transitions to a low–flow regime that minimizes noise and stiction. This two–stage scheme:

- preserves precision (via fine pressure/flow resolution at small errors),
- avoids uncontrolled pressure build–up (inner loop vents as needed), and
- maintains adequate force away from the target by allowing higher $|\Delta p|$ when the error is large.

In practice, the outer loop provides the what (the differential pressure required), while the inner loop and valve shaping provide the how (the profile—linear, exponential, or logarithmic—by which mass flow and pressure are applied). The result is a trajectory that retains bang–bang–like authority for large errors yet settles with the precision of a finely tuned proportional system near the setpoint.

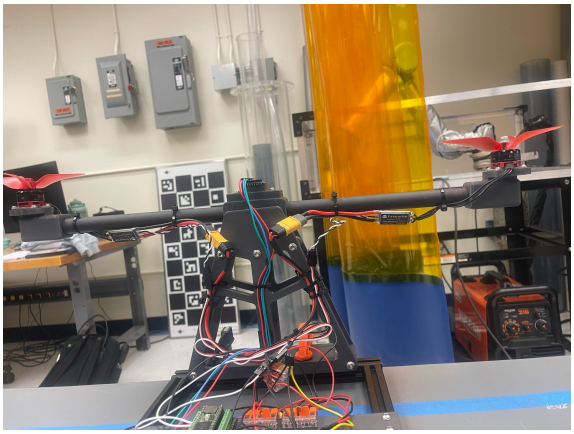

Figure 5.9: Bench-top rig for PID gain optimization with instrumented quadrotor airframe and embedded controller.

Python PID Value Optimizer (method and significance). To tune the controller systematically, a Python "PID value optimizer" and propller based testing rig, shown in Figure 5.9, that communicates with the Teensy over serial, commands a repeatable test trajectory (step and low-amplitude chirp), logs position/pressure/-valve states, and computes a scalar cost after each run. The cost combined integral absolute error (IAE) for tracking, overshoot and settling-time penalties, a valve-chatter penalty (switch count and on-time violations), and a soft penalty on control effort to discourage excessive duty. Using Bayesian optimization (Gaussian-process surrogate via scikit-optimize), the script proposed new (K_p, K_i, K_d) settings within safety bounds, wrote them to the controller, and iterated until convergence; CMA-ES was available as a fallback for non-smooth cases. This autotuning loop produced gains that were measurably better than hand tuning—shorter settling, reduced overshoot, and fewer unnecessary valve transitions—and, critically, it did so reproducibly across different loads by re-running the same experiment set. The resulting gain sets seeded the hybrid and cascade controllers, and the logged surrogate model provided a lightweight basis for gain scheduling as payload or supply pressure changed.

Proportional flow control offers excellent micro-positioning but risks cumulative pressurization without exhaust; pressure regulation eliminates that drift but weakens actuation at low setpoints. A cascaded, two-stage PID combines their strengths: inner-loop pressure authority to add/remove energy as needed, and outer-loop position control to command only as much Δp as the task requires.

5.6 Hybrid Bang–Bang with Proportional Flow Throttling Near the Setpoint

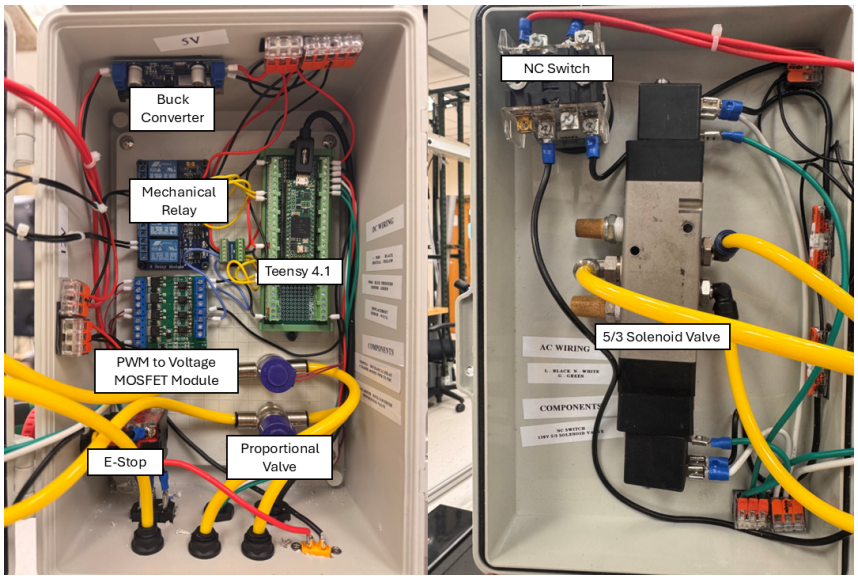

Figure 5.10: Internal layout of the PID-capable control box: Teensy 4.1, buck converter, mechanical relay, PWM-to-voltage module, e-stop, proportional valve driver, NC switch, and 5/3 solenoid valve.

Figure 5.11: External view of the PID-capable control box integrated with the modular test rig, showing sensors, NC switch, and E-stop interfaces.

The hybrid controller combines binary direction selection (CAP/ROD/HALT) with a proportional–valve PID that throttles total flow as the setpoint is approached. The PID computes a correction

Let $e = r - x$ and choose thresholds $\varepsilon_{\mathrm{BB}} > \varepsilon_{\mathrm{FLOW}} > \texttt{Buffer} > 0$. The controller operates as:

$$\text{(I) Bang–bang region:} \quad |e| \geq \varepsilon_{\mathrm{BB}}$$

$$\text{(II) Throttle region:} \quad \varepsilon_{\mathrm{FLOW}} \leq |e| < \varepsilon_{\mathrm{BB}}$$

$$\text{(III) Settle/hold:} \quad |e| < \text{Buffer}$$

A small hysteresis ($\pm\delta$) is added around $\varepsilon_{\mathrm{FLOW}}$ to prevent rapid toggling between modes.

Inside the throttle region the directional decision still selects which side "wins" (CAP for $e > 0$, ROD for $e < 0$), but we limit the chamber inflows with a low-gain PID on $|e|$. As $|e| \to$ Buffer, reducing the mass flow and, via $pV = mRT$, limiting the rate of pressure rise so that the higher-force chamber still nudges the piston but with much lower momentum. This realizes a "soft landing" while preserving the simple CAP/ROD logic.

Far from the target, unrestricted bang–bang maximizes differential pressure for speed. Near the target, unrestricted flow can pump air into both chambers faster than it can be dissipated (no exhaust), causing pressure ratcheting, harder braking, and overshoot. The throttle PID reduces $\dot{m}$ as a function of $|e|$, keeping p_c and p_r from climbing unnecessarily and cutting the kinetic energy that must be absorbed at entry to HALT. The result is shorter post-HALT oscillation and better repeatability, at the expense of a slightly slower final approach.

5.6.1 Implementation sketch (aligned with CAP/ROD/HALT).

```
double error         = setpoint - actPosition;
integral            += error * dt;
double rawDerivative = (error - previousError) / dt;

double correction    = Kp*error + Ki*integral + Kd*rawDerivative;

// Clamp to [0,255] and scale to PWM space
double limitedCorrection = constrain(correction, 0, 255);
int    scaledCorrection  = map(limitedCorrection, 0, 255, 0, 255);

// Shared throttle: higher |u| -> lower opening
int PropValveCap = 255 - abs(scaledCorrection);
int PropValveRod = 255 - abs(scaledCorrection);

previousError = error;

// Deadband: hold and close metering valves
if (fabs(error) < Buffer) {
```

```
  action = HALT;
  analogWrite(PROPORTIONAL_VALVE_CAPSIDE, 0);
  analogWrite(PROPORTIONAL_VALVE_RODSIDE, 0);
}
// Extend: CAP
else if (error > Buffer) {
  action = CAP;  // (binary directional valve logic elsewhere)
  analogWrite(PROPORTIONAL_VALVE_CAPSIDE, PropValveCap);
  analogWrite(PROPORTIONAL_VALVE_RODSIDE, PropValveRod);
}
// Retract: ROD
else { // error < -Buffer
  action = ROD;  // (binary directional valve logic elsewhere)
  analogWrite(PROPORTIONAL_VALVE_CAPSIDE, PropValveCap);
  analogWrite(PROPORTIONAL_VALVE_RODSIDE, PropValveRod);
}
```

5.6.2 Bang–Bang + PID Results and Analysis (Against Gravity)

Figures 5.12, 5.13, and 5.14 summarize the behavior of the bang–bang controller augmented with a proportional valve rule when the piston must raise a load against gravity. The position trace in Figure 5.12 shows a monotonic approach to the setpoint with essentially zero overshoot. Compared with pure bang–bang, the rise is slower but smoother; the error decays steadily to zero and remains there. The corresponding command in Figure 5.13 saturates at $u = +1$ while the error is large (maximizing upward force), then drops to the neutral/hold state ($u = 0$) once

the target is reached, avoiding the repeated reversals that produced the limit cycle previously. The pressure data in Figure 5.14 explain this outcome: cap-side pressure ramps to $\sim$ 75–80 psi while the rod side remains near $\sim$ 14–16 psi, yielding a positive differential $\Delta P \approx 55-65$ psi that is sufficient to overcome gravity and accelerate the piston upward. After the command returns to $u = 0$, the valve closes, trapping chamber pressures; the trapped ΔP balances the static weight so the actuator holds position without oscillation.

Implications. (i) The proportional augmentation reduces overshoot and eliminates the steady oscillation seen with pure bang–bang by removing the reverse-drive phase near the setpoint. (ii) The neutral hold with trapped pressure is effective for load support against gravity, but long-term accuracy will depend on leakage and temperature drift. (iii) The trade-off for smoother regulation is a longer rise time, since the controller spends more time at $u = +1$ before entering the hold state.

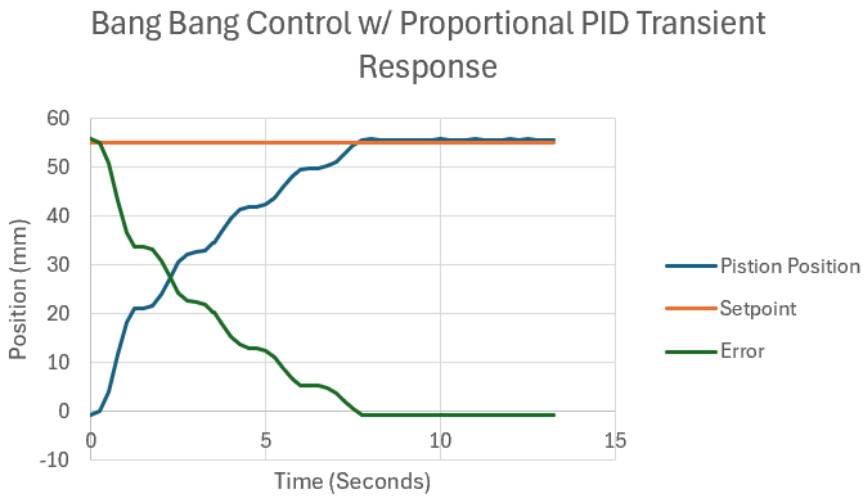

Figure 5.12: Bang–bang with proportional valve rule: piston position (blue) tracks a constant setpoint (orange) while the error (green) decays monotonically to zero with no overshoot when lifting against gravity.

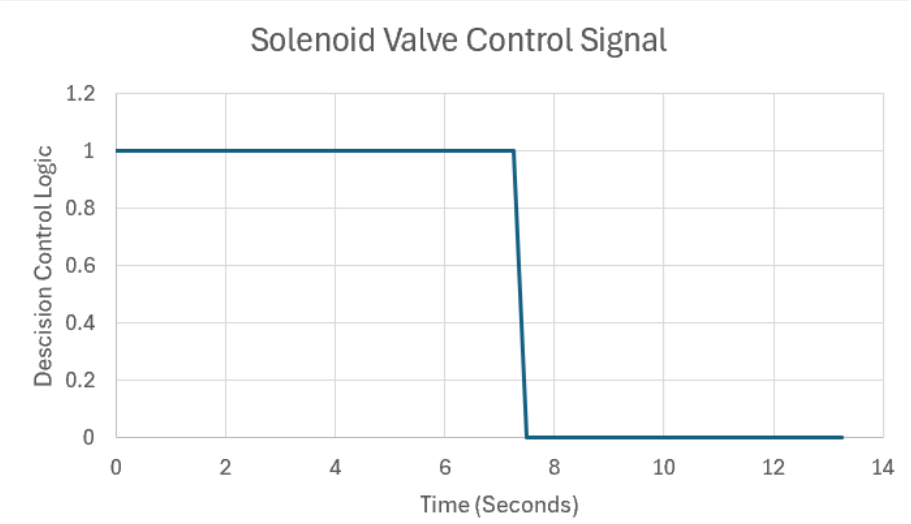

Figure 5.13: Solenoid command u: saturation at $u = +1$ during the gross lift phase, followed by a transition to neutral/hold ($u = 0$) near the target, which prevents reversals and eliminates the limit cycle at the setpoint.

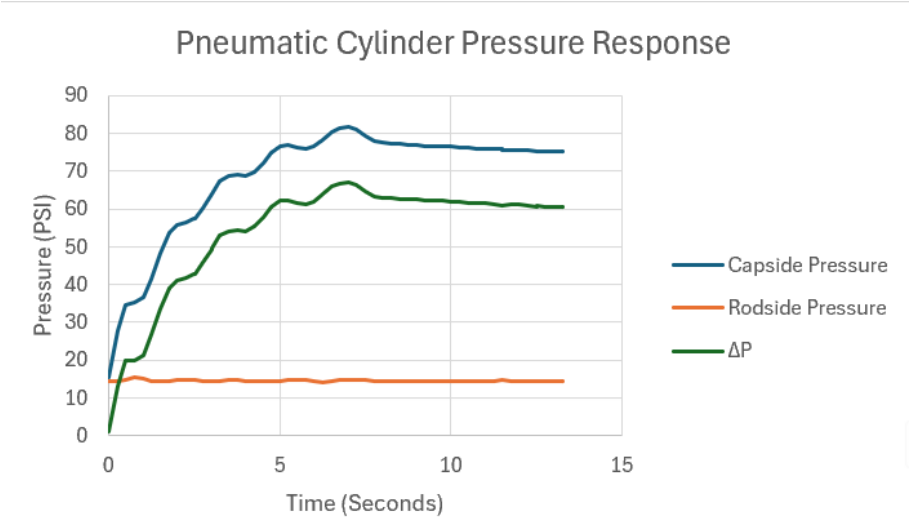

Figure 5.14: Chamber pressures and differential $\Delta P = P_{\text{cap}} - P_{\text{rod}}$ under gravitational load. A high positive ΔP drives the ascent; after the controller enters hold, trapped pressures supply the static force required to balance gravity and keep the piston at the target.

5.6.3 Bang–Bang + PID Results and Analysis (With Gravity)

Figures 5.15, 5.16, and 5.17 show the controller behavior when piston motion is aided by gravity. The position trace in Fig. 5.15 starts high and decays smoothly to the setpoint with no overshoot. While the error is large, the command holds $u = -1$ (Fig. 5.16), which vents the cap side and feeds/holds the rod side so that gravity provides most of the accelerating force. As the piston approaches the target, the command transitions to the neutral/hold state ($u = 0$), preventing a reversal and eliminating the limit cycle at the setpoint. The pressure data in Fig. 5.17 confirm this: cap-side pressure decays from ~ 65 psi toward ~ 18 psi while the rod side stays near $\sim 14-16$ psi, so the differential $\Delta P = P_{\text{cap}} - P_{\text{rod}}$ reduces to a small positive value that merely balances residual friction once the neutral state is entered.

Implications. (i) Motion aided by gravity achieves a monotonic, non-overshooting approach with lower commanded effort, improving energy efficiency. (ii) The absence of reversals near the setpoint reduces valve wear and air consumption. (iii) Long-term holding accuracy will depend on leakage and drift because the final state relies on trapped pressure in neutral.

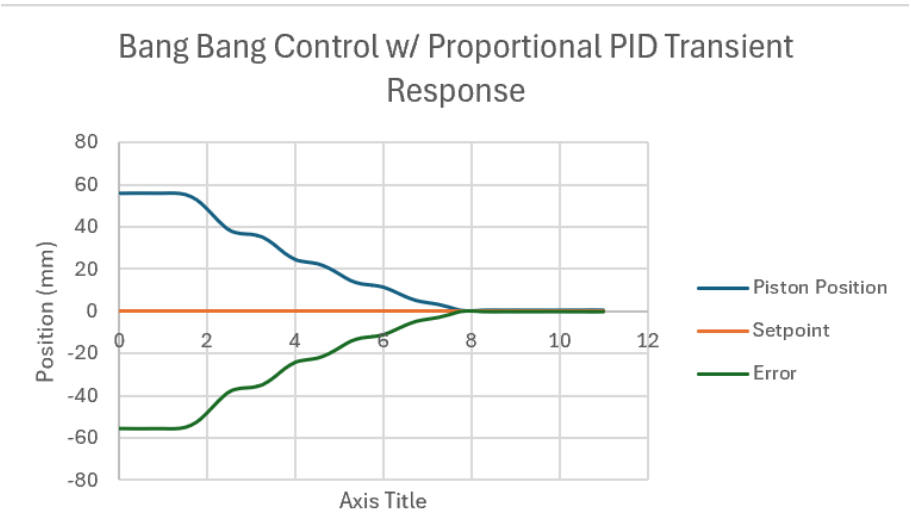

Figure 5.15: Bang–bang with proportional valve rule, motion *with* gravity: piston position (blue) approaches the setpoint (orange) monotonically; the error (green) decays to zero without overshoot.

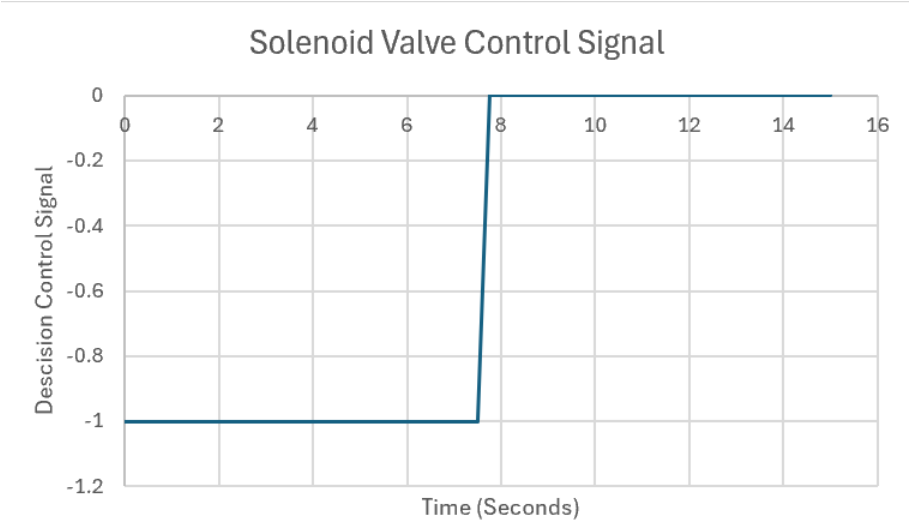

Figure 5.16: Solenoid command u for motion aided by gravity. The controller holds $u = -1$ during the descent to exploit gravitational assistance, then transitions to neutral $u = 0$ near the target to avoid reversal and chatter.

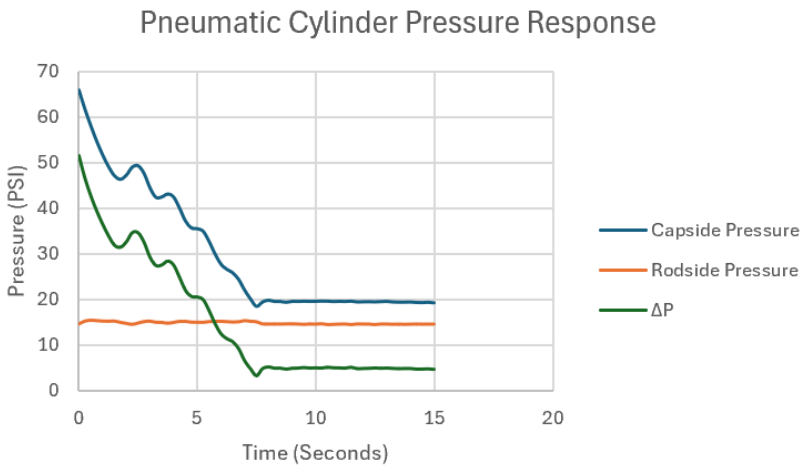

Figure 5.17: Chamber pressures and $\Delta P = P_{\text{cap}} - P_{\text{rod}}$ with gravity assisting. Capside pressure decays while rod-side remains near constant, driving a decreasing ΔP that leads to a smooth stop when the valve enters neutral and pressures are trapped.

5.6.4 PID Gain Optimization Results

Using the optimization procedure described previously in Section 5.5.1, the controller was tuned for the pneumatic stage. The observed overshoot and residual limit cycles are consistent with the pneumatic model in Chapter 3: at large errors, the supply valve operates in (effectively) choked flow, fixing the mass-flow rate and producing a high chamber pressurization slope, whereas near the setpoint the transition to unchoked flow and increased effective compliance (due to line volume and chamber compressibility) slows the dynamics and exposes friction and dead zones. The hybrid controller mitigates this by using high-duty PWM (bang–bang–like authority) to traverse the choked-flow regime quickly, then handing off to PID with pressure/velocity feedback in the unchoked regime to suppress chatter and windup. The resulting reduction in overshoot and settling time matches the predicted behavior from the identified valve orifice model and the pressure–volume relationship used in Chapter 3.

Against gravity. In the previous case (Fig. 5.12), the piston climbs in distinct slope segments and reaches the 55 mm setpoint at approximately 8 s with a small overshoot (< 1 mm) followed by weak post-settling ripples (a residual limit cycle around the deadband). In the optimized run (Fig. 5.18), the trajectory is monotonic and linear to the target (50 mm) in roughly 7–8 s, then remains flat with no observable overshoot or ringing. Both responses achieve near-zero steady-state error; the second shows a single transition into hold, whereas the first exhibits micro-corrections after reaching the setpoint. The ripple-free, monotonic response reduces valve switching and airflow—improving efficiency and hardware life—and yields tighter, more repeatable final positioning.

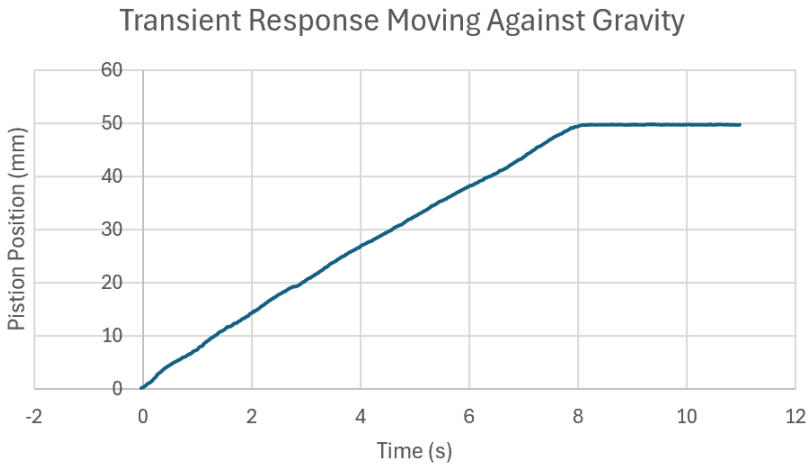

Figure 5.18: Optimized transient piston position while moving against gravity: monotonic rise to the 50 mm target with no overshoot.

With gravity. In the optimized with-gravity response (Fig. 5.19), position drops from ~ 48 mm to 0 mm in a largely monotonic fashion: an initial step near $t \approx 4$–5 s is followed by a smooth exponential-like descent to zero by ~ 17 s, with no overshoot or ringing. In the previous case (Fig. 5.15), the position rises in staged slopes and reaches the 55 mm setpoint at ~ 8 s; error decays to zero with only trace post-settling

ripple ($< 1\,\text{mm}$), indicating a small residual limit cycle around the deadband. The monotonic, ripple-free descent with gravity minimizes valve switching and air use during approach, while the tuned hybrid controller achieves a similarly fast time-to-target against gravity but with tighter final regulation and negligible overshoot.

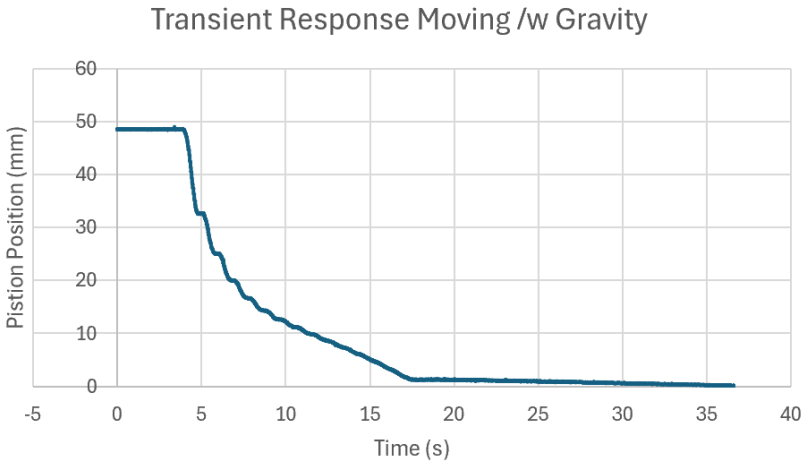

Figure 5.19: Optimized transient piston position while moving with gravity.

CHAPTER 6

CONCLUSIONS AND FUTURE WORK

This thesis addressed the technical, environmental, and operational challenges of obtaining representative post-retrieval samples from Hanford Site waste tanks by designing and validating a pneumatically actuated, articulated manipulator suitable for off-riser deployment. The work integrated (i) a physics-based pneumatic actuator model (mass flow, chamber pressure dynamics), (ii) a transmission-level coupling that maps piston motion to joint rotation under clearly stated assumptions, (iii) URDF-driven kinematic validation and physics-aware simulation, and (iv) a control strategy that progresses from on/off actuation to a hybrid scheme that adds proportional flow throttling near the target to achieve precise positioning.

On the modeling and simulation side, URDF-based visualization in RViz verified frame consistency, joint limits, and self-collision behavior prior to hardware work, while import into NVIDIA Isaac Sim exposed gravity, contact, and solver settings that affect tracking envelopes and de-risked tuning choices before bench testing. Together, these tools provided fast iteration on kinematics and revealed structural and control sensitivities (e.g., overshoot, elastic deflection), informing gain, limit, and feedforward selections for the intended duty cycle.

Control development began with bang–bang actuation to exploit maximum differential pressure for large excursions, then introduced a proportional metering stage near the setpoint. The hybrid rule produced a monotonic approach with essentially zero overshoot when lifting against gravity; the controller transitioned to a neutral/hold state that trapped chamber pressures to support the static load, at the expense of a modestly longer rise time. These results demonstrate a practical pathway to precision with compressible-flow hardware using simple embedded logic and plant-aware scheduling.

Finally, the manipulator concept and control approach were developed in the context of stringent Hanford constraints (riser access, 75 ft span coverage, deep reach, high radiation, hydrogen ignition hazard, and liquid-management limits). These constraints motivated a compact, suspended architecture with modular joints and predominantly dry, mechanically aggressive end-effectors, steering designs and controls toward inherently spark-safe actuation and robust operation in corrosive, high-humidity environments.

Future Work

- **Pressure-feedback and observers.** Augment position control with chamber-pressure feedback and state observers (e.g., extended Kalman filters) to estimate effective compliance and friction states online, enabling gain scheduling when the chain or geometry induces state-dependent transmission gains.
- **Advanced control.** Evaluate model predictive control or sliding-mode variants that blend fast approach authority with constraint handling (pressure limits, valve duty) and robustness to payload variability; compare to the demonstrated hybrid rule using common transient/steady metrics.
- **Full-system mock tank testing.** Execute end-to-end trials in a scaled tank mockup with representative obstacles, humidity, temperature, and radiation surrogates to validate reach, sample quality, and recovery procedures under the operational constraints that motivated the design.
- **Tooling and liquid management.** Broaden the end-effector suite (percussive, penetrative, and capture devices) with integrated containment/vacuum to minimize free liquid, and quantify sampling representativeness across waste

- **Parameter identification and HIL.** Perform systematic identification (valve coefficients, leakage, friction) and hardware-in-the-loop tests to tighten the fidelity of the Simulink plant used for controller tuning and to streamline field calibration.

Overall, the thesis demonstrates that a compact, pneumatically actuated manipulator—validated in URDF/RViz and physics simulation and governed by a hybrid control law—can meet the precision and safety demands of off-riser sampling while conforming to Hanford's unique constraints. The recommendations and future work above outline a direct path from laboratory validation to reliable field deployment.

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